Algorithms and Responsibility

Felix Bönisch*

May 21, 2026

Abstract

A long line of experimental work suggests that decision-makers can—and do—use delegation to shift responsibility for unwelcome outcomes onto an intermediary. We ask whether the same logic carries over to delegation to algorithms. In a preregistered online experiment ($N = 322$ UK Prolific participants, 161 in each role), a decision-maker makes a payoff-relevant prediction for a matched recipient with the option to delegate to an algorithm calibrated, at the individual level, to perform as well as she does. We manipulate, between subjects, whether the matched recipient can punish her for the decision. Two findings emerge. First, contrary to the preregistered prediction of the responsibility-avoidance literature, the *prospect* of punishment *reduces* rather than increases the likelihood of delegating to the algorithm (a 17 percentage-point gap, two-sided $p = 0.033$). Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions; the point estimate runs in the predicted direction but the test is underpowered to rule out moderate insulation effects. We discuss several candidate mechanisms for the headline reversal and argue that, among those we can evaluate, a process-ownership account—in which the moral salience introduced by the punishment possibility raises the value of being the proximate cause of a good outcome—is the most consistent with the data, while flagging that the supporting evidence is exploratory and the design cannot rule out experimenter demand. Within the bounds of the setting we study, our results offer a behavioral complement to regulatory frameworks that hold human users accountable for algorithm-mediated decisions: such frameworks may discipline algorithmic adoption rather than merely allocate post-hoc responsibility.

JEL Classification: C91, D81, D91, O33

Keywords: algorithms; delegation; responsibility; punishment; experiment

*WZB Berlin Social Science Center. Email: `felix.boenisch@wzb.eu`. Acknowledgements to be added at submission.

1 Introduction

Algorithms are increasingly involved in consequential decisions: they screen job candidates, set credit scores, recommend medical treatments, evaluate parole applicants, and—more mundanely but at vast scale—route the cars driving toward this very page. As the locus of decision-making shifts toward algorithmic systems, a parallel question has moved into public, legal, and academic debate: when an algorithm-mediated decision turns out badly, who bears the blame? A natural intuition, which finds support in a long line of experimental work on responsibility avoidance through delegation, is that decision-makers can use algorithms as a means to shift blame onto a non-human intermediary. To return to a familiar example: if you and a friend arrive late because she relied on a navigation app rather than her own sense of direction, the algorithm seems a convenient scapegoat—and the next time she is unsure of the route, the prospect of avoiding your blame may itself motivate her use of the app. This paper provides experimental evidence on whether either of these intuitions is borne out.

We address two questions. *First*: can decision-makers actually shift responsibility away from themselves by delegating to an algorithm—that is, are they punished less for a delegated decision than for a self-made decision with the same outcome? *Second*: does the prospect of being held responsible motivate the decision to delegate in the first place? The two questions correspond to, respectively, the *viability* and the *motivational salience* of algorithmic responsibility avoidance. They have been studied in different settings and rarely in combination, and previous designs have struggled to identify them cleanly. We do so here in a preregistered online experiment.

Our experimental design pins down both questions. Player A first completes ten rounds of an incentivised visual prediction task (estimating a person's weight from a facial image and stated height). She then makes an final prediction whose outcome determines the payoff of a second participant, Player B: a high payoff if the prediction falls within ten pounds of the true weight, a low payoff otherwise. Player A may either make this prediction herself or delegate it to an algorithm. The algorithm is calibrated to mimic Player A's individual performance from the first ten rounds—that is, on the final prediction it produces the high outcome with the same probability as Player A herself. Player A is told this. We manipulate the punishment possibility between subjects: in the *Punishment* condition, Player B may reduce Player A's payoff conditional on the delegation decision and the realized outcome; in the *No-Punishment* condition, no punishment is possible. Player B's punishment behavior is elicited via the strategy method across all four (delegation, outcome) cells, providing a within-subject test of whether delegation insulates Player A from punishment.

Two design features distinguish our setting from the closest precedent, Feier et al. (2022). First, the algorithm is calibrated at the *individual* level rather than the session or group level. Existing designs typically expose subjects to a distribution of algorithmic performance and let them infer their own ability relative to that distribution; the delegation decision then carries strong signal value about the principal's self-perception, and any differential punishment can reflect overconfidence rather than responsibility shifting per se. By neutralizing relative-performance signaling at the individual level, we isolate the responsibility-shifting motive. Second, our *No-Punishment* condition turns off the punishment possibility entirely, allowing us to identify the effect of anticipated punishment on the delegation decision itself—a test that is not feasible in designs where punishment is always possible. Both features sharpen the test of responsibility shifting in algorithmic delegation that the literature has thus far been able to deliver.

Two findings emerge, and in their joint structure they push against the natural reading of the responsibility-avoidance literature. *First, the prospect of punishment reduces, rather than increases, the likelihood of delegation:* 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not (a 17 percentage-point gap; two-sided Pearson Chi-squared $p = 0.033$, Fisher exact $p = 0.041$; the preregistered one-sided test in the H1 direction is non-significant by construction, since the result reverses the predicted direction—we discuss this in detail in Section 4). This rejection of the standard prediction in favor of the opposite direction is the paper's main empirical contribution, and we view it as a behavioral counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms. *Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions in our setting:* Player B's punishment of Player A is statistically indistinguishable across the two, and Player A correctly anticipates this absence of differential punishment in her elicited beliefs. The point estimate of any insulation effect runs in the predicted direction but the test is underpowered against moderate alternatives ($n \approx 67$ Player Bs after attention-check exclusion); the broader claim that responsibility shifting through algorithmic delegation is altogether absent is one we cannot make with this data.

What mechanism reconciles the two findings? Among the candidates we can evaluate with our design, a *process-ownership* account is the most consistent with the data—though we treat it as a candidate hypothesis rather than an established mechanism, since the supporting evidence is exploratory and a fourth candidate (experimenter demand) cannot be ruled out by construction. A simple anticipation story—in which Player A delegates less when punishment is on the table because she expects harsher punishment for delegated decisions—is inconsistent with the elicited beliefs and with Player B's actual choices. A simple effort story—in which Player A makes the prediction herself in order to outperform the algorithm and earn the high payoff—is inconsistent with the data, in which non-delegators in the *No-Punishment* condition outperform non-delegators in the *Punishment*. What remains is the milder hypothesis that, when the moral salience of the decision is heightened by the punishment possibility, Player A wants to be the proximate cause of any good outcome for Player B, irrespective of any quantifiable belief about how she will be judged. Consistent with this interpretation, the negative correlation between Player A's task performance and her propensity to delegate is observed in the *Punishment* condition only: those most likely to deliver the high payoff want to do so personally when the moral stakes are made salient. Section 4.1 weighs the four candidate mechanisms in detail and is explicit about what the data can and cannot rule out.

These findings contribute to three strands of literature. First, we add to the experimental literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016), by extending the canonical setting to delegation to an algorithmic intermediary in a setting with genuine outcome uncertainty. Second, we contribute to the rapidly growing literature on algorithm aversion and adoption (Dietvorst et al., 2015; Logg et al., 2019; Burton et al., 2020; Mahmud et al., 2022; Chugunova and Sele, 2022) by isolating an under-studied determinant of algorithm use: the institutional context in which the human user is held accountable. Third, and most directly, we build on Feier et al. (2022) and Kirchkamp and Strobel (2019) by providing a clean experimental test of two long-standing claims in the responsibility-and-algorithms literature, with a design that controls relative-performance signaling at the individual level and that includes a condition in which punishment is ruled out by construction. The contrast with Feier et al. (2022) is consequential: that study finds that decision-makers are held *less* responsible for algorithmically delegated decisions, while we find that the punishment possibility *reduces* delegation rather than increasing it. We argue in Section 2 that the two findings are reconcilable: the relative-performance signaling that drives much of Feier et al.'s differential punishment is closed off by construction in our design, isolating a different and milder margin (the motivational salience of accountability for the principal). Beyond the experimental literature, our findings speak to the regulatory debate over the assignment of responsibility for algorithmic decisions (European Union, 2024): within the bounds of the setting we study, holding human users accountable for algorithm-mediated outcomes appears to be a behavioral lever for direct human engagement with the decision, rather than merely a tool for ex-post moral attribution. We are explicit in Section 5 about why the present evidence—a single online experiment on a stylised prediction task with one UK Prolific sample—supports this claim only in a bounded sense, and what would be required to extend it to the high-stakes algorithmic domains (medical, judicial, hiring, financial) that motivate the regulatory debate.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature in detail. Section 3 describes the experimental design and states our two preregistered hypotheses. Section 4 presents the empirical findings, including the mechanism analysis. Section 5 discusses the implications and limitations of our results.

2 Related Literature

This paper bridges three strands of literature: (i) responsibility avoidance through delegation, an established line of experimental work in which the intermediary is another human; (ii) algorithm use, which examines when and why people are willing to use algorithmic decision aids, for themselves and others; and (iii) responsibility attributions to decisions involving algorithms, an emerging literature at the intersection of the first two.

[Comment: Is this first paragraph really necessary? Is it perhaps better placed in a conclusionary paragraph of the whole literature section, perhaps introducing the summary of the contribution of the paper?]

Responsibility avoidance through delegation

Delegation has long been studied as an efficiency-enhancing institutional arrangement: principals delegate to agents who have better information, lower opportunity costs, or specialised skills. Alongside these efficiency motives, however, a separate literature has identified a strategic responsibility-shifting motive — principals may delegate not merely because the agent is more competent or cheaper, but because doing so shifts the responsibility for the resulting outcome away from the principal — both in the principal's own eyes (her felt responsibility for the outcome) and in the eyes of others (the blame that others attribute to her). [Comment: Are there any examples of ancient philosophers?] The political-science and public-choice literatures have long recognized that delegation can serve as a strategy for shifting blame onto an intermediary (Weaver, 1986; Fiorina, 1986). Experimental economics has established the same point in incentivised settings, primarily using modified dictator games.

Bartling and Fischbacher (2012) provide the canonical experimental evidence. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to an intermediary whose monetary incentives are aligned with the dictator's. The recipient is adversely affected if the unfair allocation is chosen and may punish the dictator and the intermediary. They find that through delegation punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. Hamman et al. (2010) extend this picture in a setting without third-party punishment. Still, principals' transfers to recipients fall sharply when they delegate to an agent, indicating that the responsibility-shifting motive here must operate rather through felt responsibility or moral wiggle room. Self-reported responsibility and recipient-side blame attributions are consistent with this reading—dictators feel less responsible when they delegate, and recipients place less blame on principals than on intermediaries.

Coffman (2011) reinterprets the mechanism behind this punishment reduction. He finds that the principal can no longer avoid punishment when she directly interacts with the receiver despite the intermediary's presence, suggesting that what reduces punishment is the absence of direct interaction with the affected party rather than any transfer of blame to a deciding intermediary. This holds even when the intermediary remains completely passive. Oexl and Grossman (2013) sharpen this reading by modifying the canonical game so that delegation necessarily eliminates the fair option. The principal nevertheless escapes punishment, with the intermediary actively choosing between two unfair allocations after delegation. Finally, both Bartling and Fischbacher (2012) (with a die roll) and Oexl and Grossman (2013) (with an intermediary-triggered random draw) include treatments that substitute a randomization device for the human intermediary. In both, the principal still shifts some punishment to the device, though less than under delegation to a deciding human. Across the two papers, two distinct components of the responsibility-shifting effect emerge — the absence of direct interaction between principal and recipient drives the reduction in principal punishment, while the act of choosing, rather than any contribution to outcome unfairness, is what transfers blame to the intermediary.

Beyond the canonical dictator-game setting, the responsibility-shifting motive — and the strategic use of delegation it gives rise to — has been documented across a range of domains. In incentivised ultimatum bargaining, proposers using a hired delegate extract larger surpluses (Fershtman and Gneezy, 2001). In sequential voting, non-pivotal voters are punished less than pivotal ones for the same group outcome and voters strategically vote against their preferred allocation to avoid being pivotal (Bartling et al., 2015). For acts of deception, principals pay an agent to lie on their behalf (Erat, 2013). In policy-making, observers rate delegating legislatures as less blameworthy than legislatures who decide directly, even when the delegate is powerless to change the outcome (Hill, 2015). Freer et al. (2024) isolate the motive in financial decision-making, finding that a substantial fraction of investors delegate even purely luck-based lottery choices, where alternative motives — decision costs, performance-chasing, or risk tolerance — cannot rationalize the choice.

In sum, the canonical literature documents that in settings with self-interested choice delegation reduces principal punishment and her felt responsibility, in turn motivating delegation. Our setting departs from this canonical design along three dimensions: the intermediary is algorithmic rather than human, the outcome falls entirely on a third party rather than partly on the principal herself, and the underlying decision involves genuine predictive uncertainty rather than a known fair/unfair choice. Whether the responsibility-shifting motive retains its force under these conditions is the question we take up.

Delegation to algorithms

Since we are interested in whether similar motives of responsibility avoidance also exist when the intermediary is algorithmic, our work also relates to the rapidly growing field of prefernences for algorithm use in decision-making. The literature on people's willingness to use algorithmic decision aids has grown enourmously rapidly since Dietvorst et al. (2015), who introduced the term *algorithm aversion* for the finding that people prefer their own (or other humans') judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers. In contrast, Logg et al. (2019) document conditions under which subjects show *appreciation* of algorithmic advice over human advice; the picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Burton et al. (2020), Mahmud et al. (2022), and Chugunova and Sele (2022) provide systematic reviews of this literature and document a large array of moderators for when and why algorithms are used in decision-making. For example, algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes (Berger et al., 2021), when the algorithm's reasoning is opaque (Litterscheidt and Streich, 2020; Sharan and Romano, 2020), when the algorithm is presented as more complex (Stein et al., 2020). Algorithm aversion is also stronger when the task is perceived as more uncertain (Dietvorst and Bharti, 2020), when delegation removes any ability to override the algorithm's choice (Dietvorst et al., 2018), and when the algorithm is used in tasks perceived as subjective (Castelo et al., 2019; Bogert et al., 2021). Recent additions include Bockstedt and Buckman (2025), who show that loss-framing of decision tasks eliminates the preference for human assistance over AI assistance, and Ivanova-Stenzel and Tolksdorf (2024), who develop a menu-based design to elicit willingness to cede control to algorithms versus humans.

[Comment: - the literature reviews and references in general are old-ish...ideally we can include more recent references. - Maybe include the Dargies, Hakimov, Kübler reference here. - Also think about where to include the two ivanova stenzel papers]

Responsibility perceptions of decisions involving algorithms

Within this body of work, a smaller subset focuses on algorithm-use preferences in settings whose consequences fall partly or wholly on someone other than the decision-maker, sometimes called the *moral domain* (Gogoll and Uhl, 2018; Bigman and Gray, 2018; Jauernig et al., 2022). The consensus across this subset is that aversion to algorithms is amplified once third-party welfare is at stake: Gogoll and Uhl (2018) document this pattern in an incentivised calculation task affecting a partner — a straightforward arithmetic task where algorithmic advantage should have been salient.; Bigman and Gray (2018) extend it to high-stakes moral domains (driving, legal, medical, military); Jauernig et al. (2022) identify *moral discretion* as the human capacity people specifically want preserved.

With the decisions affecting others, the *moral domain* in delegating to algorithms also introduces the responsibility-shifting motive that characterizes setting of delegation to humans.

Qualitative (unincentivized) edvidence, which examines how third parties attribute responsibility and blame to human actors involved with algorithms actually goes way back: Naquin and Kurtzberg (2004) find that observers attribute less responsibility to a company when an accident involves technology; Pezzo and Pezzo (2006) document the same pattern for a doctor who follows a (bad) recommendation from an automated decision aid; and Liu (2021) replicate the effect in a contemporary setting. Shank et al. (2019) consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine. Bigman and Gray (2018) document a robust "algorithmic outrage asymmetry": moral violations by algorithms produce less outrage than identical violations by humans. Tobia et al. (2021) extend the discussion to liability law and find experimentally that physicians who follow standard-care AI recommendations face reduced legal exposure; they argue that this could constitute an incentive to follow such recommendations specifically because doing so shields physicians from blame. The consistent message across this strand is that responsibility attributions to humans involved with algorithms differ systematically from attributions to humans acting alone, but the direction and magnitude depend on the precise configuration.

Evidence using incentivised laboratory experiments with monetary stakes — is younger and smaller. Kirchkamp and Strobel (2019) compare a modified dictator game in which the dictator either acts alone or shares responsibility with a computer; they find no significant differences in self-reported responsibility between human-computer and human-human teams, although a (statistically insignificant) tendency towards more selfish allocations under shared responsibility with humans. Maasland and Weißmüller (2022) find in a computer-aided HR setting that algorithm aversion *dominates* blame avoidance: the prospect of using an algorithm to delegate an unpleasant HR decision (a dismissal) does not overcome the underlying aversion to algorithms in this domain. The most closely related study is Feier et al. (2022), who examine delegation to an algorithm in an incentivised laboratory experiment. In their design, subjects first complete ten logic tasks, are randomly assigned to either a "human" or "machine" treatment, and then either rely on their own performance or delegate to (respectively) another human or an algorithm to determine a third party's payoff. Third parties may then reward or punish. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome.

The Feier et al. (2022) design, which is the immediate precedent for ours, has three features that complicate the inference about pure responsibility shifting. First, the algorithm in their experiment is calibrated at the session level rather than at the individual level: each subject is shown the distribution of algorithmic performance across the relevant pool, and her delegation decision then reflects beliefs about her own ability *relative* to that distribution. The delegation choice therefore carries strong signal value about Player A's self-perception, which contaminates downstream punishment as a measure of responsibility shifting per se; Feier et al. (2022) indeed report that the delegation decision is well explained by perceived own performance relative to the algorithm. Second, the same subject acts as both delegator and evaluator, introducing additional motives — self-justification, hypocrisy aversion, projection — into the punishment decision. Third, their design does not include a treatment in which punishment is ruled out by construction, so the question of whether the punishment possibility itself motivates delegation cannot be addressed in their data.

Four recent studies investigate algorihm use and responsibility assignment further. Chevrier and Teixeira (2024) extend the blame chain to include the programmer behind the algorithm: in their design, AI programmers and the delegator can both escape responsibility for inegalitarian AI-mediated allocations, creating a "moral wiggle room" that incentivises programmers to push the AI toward more inegalitarian choices. Köbis et al. (2025), in a high-profile study in *Nature*, document that humans request unethical behaviour from machine agents more readily than from human agents, in part because the machines (large language models including GPT-4, Claude 3.5 Sonnet, and Llama 3.3) comply with such instructions 58–98% of the time, compared to 25–40% for human agents. Tontrup and Sprigman (2025) examine delegation of a dictator-game allocation to ChatGPT and find that delegators transfer less to the recipient than non-delegators, indicating that AI delegation is used strategically to extract surplus while shedding moral responsibility. Hüholt and Szech (2026) use an incentivised trolley-style donation choice between two real charitable allocations in two online experiments and find that subjects delegate the moral decision to an AI significantly more often than to a human counterpart. Although these four settings differ (active dishonesty under instruction, dictator-game allocation, and charitable donation), the AI-as-blame-shield motif is shared, and the direction of the effect on delegation is uniformly positive — in contrast to Maasland and Weißmüller (2022), where algorithm aversion dominates the responsibility-shifting motive in the same broad domain.

[Comment: Should we say more explicitely that this more recent work has documented the opposite pattern as earlier work mentioned above?]

[Comment: For the transistion we should say that the relative performance of algorithms has always driven the delegation decision or at least confounded the interpretation of any responsibility effect (Is that so? Do the papers explicitely state this?)]

Where this paper fits

Our experiment is the first to our knowledge to (i) calibrate the algorithm to mimic each individual subject's performance, eliminating the relative-performance confound; (ii) separate the role of delegator and evaluator across distinct subject pools; and (iii) include a between-subject manipulation that turns the punishment possibility on or off, providing a clean test of whether anticipated punishment is a motive in the delegation decision itself. The first two features sharpen the test of the responsibility-shifting hypothesis (H2) by removing alternative interpretations of differential punishment. The third moves beyond the existing literature, which has documented punishment differentials *conditional on* the punishment possibility but has not directly identified whether the threat of punishment causally moves the delegation decision.

Reconciling our findings with Feier et al. (2022). Two of our results sit in apparent tension with the closest precedent. First, Feier et al. (2022) document that decision-makers are punished *less* for algorithmically delegated decisions; we find no significant punishment differential. Second, the natural prediction from Feier et al. (2022)'s own findings is that the prospect of punishment should make algorithmic delegation *more* attractive; we find the reverse. We do not view these contrasts as contradictions, but as evidence that the design features which differ between the two studies are doing real work. The differential punishment in Feier et al. (2022) is most plausibly read as third parties extracting a signal from the delegation decision about the principal's self-perceived ability relative to the algorithm—a signal that is maximally informative when (as in their design) the algorithm is calibrated at the session level and when the principal's relative ability is uncertain. Our individual-level calibration, made common knowledge, closes off this signal-extraction channel by construction: a Player B in our setting has no basis to read delegation as overconfidence, laziness, or ability avoidance. The disappearance of the punishment differential in our data is therefore consistent with, rather than refutes, the signal-extraction reading of Feier et al. (2022)'s result. By the same logic, the responsibility-shifting motive that Feier et al. (2022)'s setup leaves on the table is muted in ours, isolating a different and milder margin: the moral salience of accountability for the principal herself, independent of any third-party inference. The reversal we observe in the delegation decision speaks to that margin—and we are explicit that the question of which design feature is causally responsible for the sign reversal cannot be resolved without an experiment that varies relative-performance calibration directly, which we do not do here.

We see scope, then, for the two studies to be parts of a single picture: Feier et al. (2022) identifies the signal-extraction channel through which delegation can shift third-party blame; we identify a process-ownership channel through which the prospect of accountability can pull the principal toward direct engagement with the decision. The two channels are not mutually exclusive. Chevrier and Teixeira (2024), looking at a third margin (the programr behind the algorithm), and Köbis et al. (2025), looking at a fourth (the compliance asymmetry between human and machine agents on instructed unethical tasks), populate further nodes in this expanding map. A unified design that varies the relative-performance calibration, the punishment possibility, and the visibility of the algorithm's compliance properties would be the natural next step. We discuss the implications for future work in Section 5.

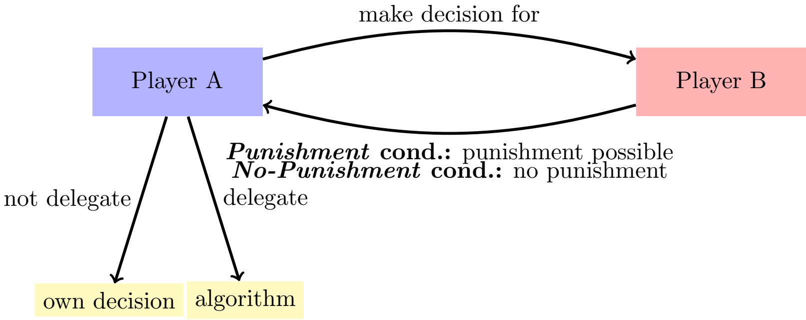
3 Experimental Design

We conducted an online experiment, preregistered at AsPredicted #133980 (aspredicted.org/hs9az8.pdf), on the British platform Prolific between February and June 2023, recruiting 322 participants from the United Kingdom. The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior.

Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B directly, or to delegate the prediction to an algorithm calibrated to perform as well as she does. Player B is the matched recipient whose payoff is fully determined by the outcome of Player A's prediction; in the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure 1 summarizes the basic structure.

We targeted a balanced design of 80 participants per role per condition, for a planned sample

Figure 1: Conditions overview



of 320 subjects.[1] The experiment was programmed in oTree (Chen et al., 2016) and run via Prolific. Eligibility followed Prolific's standard quality screens—minimum 95% prior approval rate, no warnings on file, and a fluent-English filter—together with a desktop-only restriction (no mobile or tablet) to keep the screen layout consistent across subjects.

Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. In each round, she predicted a person's weight from a facial image and the person's height. Predictions could be entered via slider or typed directly in kilograms, pounds, or stones-and-pounds; no feedback was provided between rounds.[2] At the end of the experiment, one of the ten rounds was randomly selected for payment: Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: they established a stake from which any later punishment could be deducted; they familiarized Player A with the task; and they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is deliberate. Algorithm preferences depend sensitively on the algorithm's perceived relative competence (Dietvorst et al., 2015; Logg et al., 2019);

FB CHECK: Verify that Dietvorst (2015) and Logg (2019) support the "perceived relative competence" framing. Dietvorst (2015) is largely about confidence loss after observing errors rather than about ex-ante perceived competence; Logg (2019) is about algorithm appreciation as a function of perceived ability.

as described below, our design neutralizes this concern by calibrating the algorithm to match each subject's individual performance. The visual nature of the task makes algorithmic error plausible and lends credibility to the equal-performance claim, in contrast to purely mathematical tasks where subjects may be skeptical that an algorithm could ever fail (Gogoll and Uhl,

[1] Appendix C reports the full ex-ante minimum-detectable-effect calculations underlying this cell size.

[2] All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

2018).

FB CHECK: Verify that Gogoll & Uhl (2018) actually support the math-task scepticism contrast. The paper's main claim is about moral-domain aversion; the math-task framing may not be in the paper.

Predicting body weight from a facial image has been used in earlier experiments (Logg et al., 2019) and arguably levels the playing field between human and algorithm: humans have a lifetime of practice at this mapping, while algorithms cannot easily exploit large training corpora here. Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This reflects the nature of many real-world applications of algorithms—financial forecasting, medical diagnosis, fraud detection—in which outcomes are inherently uncertain and performance is evaluated ex post.

After the ten rounds, Player A was randomly paired with another participant, Player B. Player A was informed that her final prediction would not affect her own payoff but would almost entirely determine the payoff of Player B, who had not completed the prediction task himself: if the prediction fell within 10 lbs of the true weight, Player B would receive a high payoff (£5); otherwise a low payoff (£1). Player A then learned that she could either make the prediction herself or delegate it to an algorithm calibrated to mimic her own performance from the first ten rounds, so that on expectation the algorithm would produce the high payoff for Player B at the same rate as Player A would.

Concretely, the algorithm draws a Bernoulli outcome with success probability equal to Player A's empirical accuracy rate over her first ten rounds. A subject who scored 3/10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability. This individual-level Bernoulli calibration is the central design innovation relative to the closest precedent (Feier et al., 2022), and it does substantive work in the identification argument. First, it ensures the algorithm's success probability equals Player A's empirical accuracy over the first ten rounds for every subject, making the equal-performance claim true in the data on which the calibration is based. Second, it strips any signal value that the delegation decision might otherwise carry about Player A's beliefs concerning her own ability relative to the algorithm: a Player B observing the delegation decision has no basis on which to read it as overconfident, lazy, or selfish. This closes off the most plausible third-party signal-extraction channel through which delegation might otherwise insulate the principal from punishment.

The contrast with session-level calibration is consequential. Had the algorithm instead been calibrated to reflect aggregate performance across a pool of participants—as in Feier et al. (2022)—the delegation choice would be confounded by each subject's beliefs about her own ability relative to the pool. Indeed, Feier et al. (2022) report that the delegation decision in their setting is largely driven by perceived own performance relative to the algorithm. Pairing the algorithm to each subject individually, and making this common knowledge between Player A and Player B, strips this confound out by construction.

In the *Punishment* condition, Player A was further informed that Player B could reduce her payoff by up to £2, with the deduction depending on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no punishment option. The two delegation options (own decision / algorithm) were presented in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A's intrinsic preference over delegation in the *No-Punishment* condition, since any positive cost would conflate the responsibility-shifting motive with a price effect; costless delegation is standard in the responsibility-shifting literature for this reason (Bartling and Fischbacher, 2012; Coffman, 2011; Feier et al., 2022).

We then asked Player A to report her beliefs about Player B's punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). Beliefs were elicited on the same scale as actual punishment (£0–£2 in £0.10 increments) and the four scenarios were presented in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging behavior. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen.[3] In the *No-Punishment* condition, no punishment was possible; we elicited beliefs about hypothetical punishment to maintain consistency across treatments.

Player A was then asked to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10. We elicited this belief after the delegation decision, and after Player A had learned of the equal-performance calibration, for three reasons: the post-update belief is the analytically relevant input for any subsequent behavior; pre-decision elicitation would risk introducing measurement-induced anchoring on the delegation decision itself; and the elicitation is used as a moderator rather than as a determinant of delegation in the primary analysis. The elicitation was incentivized: Player A earned an additional £0.30 with the probability she had assigned to her actual score.[4] Player A was also asked to estimate the population distribution of scores by assigning a share to each possible score from 0 to 10, which lets us construct a measure of perceived relative ability that the prior literature's pooled-calibration designs cannot identify. One score was randomly drawn at the end of the experiment and Player A earned £0.30 if her estimate for that score was within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of Holt and Laury (2002) and administered a short questionnaire on socio-demographic characteristics

[3] Subjects who failed the attention check are excluded from belief-related analyses.

[4] The mechanism is incentive-compatible only for risk-neutral subjects; in practice most subjects spread probability mass across several scores.

and technology affinity, before presenting Player A with her preliminary payoff. Figure 10 in Appendix F summarizes the experimental timeline for Player A.

Player B

After providing informed consent, Player B was told that he had been randomly matched with a Player A and was introduced to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A's final prediction—£5 if it fell within 10 lbs of the true weight and £1 otherwise—while Player A's payoff depended solely on her performance in the initial ten rounds. Crucially, Player B was also told that Player A could delegate the final prediction to an algorithm calibrated to perform equally well as Player A in the earlier rounds, and that Player A herself was aware of this equal-performance calibration. The equal-performance claim is therefore common knowledge between the two players, a feature on which our identification argument rests.

In the *Punishment* condition, Player B could choose to reduce Player A's payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through sanction-imposing behavior (Bartling and Fischbacher, 2012; Coffman, 2011; Steffel et al., 2016), and because the applied policy questions about algorithmic accountability that motivate our paper are likewise framed in sanction-language. Imposing strictly positive punishment in any cell required Player B to forgo £0.10 of his own earnings for that cell, ensuring that punishment was costly to the punisher.[5] We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B) via the strategy method, implementing only the decision matching the realized combination; the strategy method gives us the within-subject power to detect cell-level differences with the achieved sample size.[6] Because the punishment decision is the central outcome variable, we included an attention check on the punishment-elicitation screen.[7] In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B's perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before he was presented with his final payoff. Figure 11 in Appendix F summarizes the experimental timeline for Player B.

Once Player B's decisions had been recorded, Player A's final payoff was calculated and all participants were paid on the day of participation. Figure 9 in Appendix E reports the experimental flow of participants by condition and role, including recruitment, attention-check exclusion, and the analysis sample.

[5] The cost is fixed per scenario rather than proportional to the punishment level, so that strictly positive punishment is always costly relative to no punishment but the cost does not increase with the severity of the sanction.

[6] The strategy method may yield different absolute punishment levels than direct elicitation; we discuss the implications and a possible direct-method companion as future work in Section 5.

[7] The check asked the subject to confirm a numerical aspect of the payoff structure that had been displayed on the previous screen. Subjects failing this check are excluded from punishment-related analyses.

Hypotheses and identification

Our design tests two preregistered hypotheses, both grounded in the literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016) and on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018).

H1 (Delegation): The introduction of potential punishment increases the likelihood that Player A delegates the decision to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) also show that subjects are more likely to delegate morally consequential decisions even to a die roll—an early form of delegation to a non-intentional agent—suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates is less settled. Feier et al. (2022) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the *kind* of intermediary may matter less than the mere availability of one—but their design does not directly compare delegation rates across punishment regimes, leaving open the question H1 addresses.

FB CHECK: Felix asked whether Feier et al. (2022) report whether delegation propensity is HIGHER under punishment than without for both AI and human delegates. Their design does not appear to directly run this between-treatment comparison, but please verify in the paper.

H1 is identified by the between-subject comparison of Player A's delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

H2 (Punishment): Conditional on a low-payoff outcome for Player B, Player A is punished less when she delegated the decision to the algorithm than when she made it herself.

H2 mirrors H1 on the punishment side. Bartling and Fischbacher (2012) find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. Feier et al. (2022) find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Their study differs from ours in three respects, however: a different task; an emphasis on reward differentials rather than punishment; and—most consequentially—a design that does not control for relative performance at the individual level. As a result, in their setting the delegation decision carries a strong signal about Player A's beliefs concerning her own ability relative to the algorithm; a failure to delegate may then be punished as overconfidence rather than as direct responsibility-taking, contaminating the inference about responsibility-shifting per se. H2 is identified by the within-subject comparison across the four (delegation, outcome) cells of Player B's strategy-method punishment, which isolates the conditional-on-outcome effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B's punishment proclivity.

Two further design features deserve note. First, although the prediction task affects Player B's payoff, the incentives of the two players are nominally aligned: Player A has no direct financial interest in Player B receiving the low payoff, and a successful prediction strictly weakens her own expected punishment.[8] This alignment mirrors canonical real-world settings for algorithmic delegation—medical diagnosis, financial forecasting, judicial decision-support—in which the decision-maker has no direct material interest in the affected party's outcome but bears reputational or moral consequences for poor outcomes; by replicating this alignment, our design speaks directly to the policy-relevant cases. A standard intention-based punishment motive should therefore have less bite here than in dictator-game settings where the decision-maker can choose to be selfish. Second, although the underlying decision (a prediction under uncertainty) is not itself moral, the setup nonetheless qualifies as part of the *moral domain* as defined in the algorithm-aversion literature (Gogoll and Uhl, 2018): Player A's choice imposes monetary externalities on a matched recipient, and a poor decision causes real, albeit modest, monetary harm. The implications of these two design choices for the interpretation of our results are taken up in Section 4.

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with participation restricted to UK residents.[9] A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).[10] On average, Player A spent 12 minutes completing the experiment and earned on average £2.49 [Comment: Check - these numbers do not include bonus fee, etc.], while Player B spent approximately 8 minutes and earned £2.30 [Comment: Check - these numbers do not include bonus fee, etc.] on average. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix A.1, Table 4).

[8]In the *Punishment* condition, a better outcome for Player B reduces expected punishment for Player A; in the *No-Punishment* condition there is no punishment to reduce. Either way, Player A's incentives are weakly aligned with Player B's.

[9]Comment: insert exclusion criteria - To ensure all subjects perceived the experimental interface identically, phone and tablet users were not allowed to participate.

[10]We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *No Punishment* [Comment: Check] condition.

Player A – Delegation behavior

Headline. We begin by examining the delegation behavior of Player A. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the decision to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. Figure 2 illustrates the difference. The introduction of potential punishment thus significantly *reduces* the likelihood of delegation to the algorithm (Pearson Chi-squared, $p = 0.033$; Fisher's exact, $p = 0.041$) [Comment: alternative phrasing - We reject the null of equal delegation rates across conditions at the 5% level.][11] [Comment: P-values refer to Chi-squared test without Yates correction. I think we should consider using the Yates correction.] [Comment: Those are the p-values for the two-sided test. In the design section, we have one-sided hypotheses? Should we instead have equal delegation as the hypothesis and then reject that, particularly because the effect goes in the opposite direction as hypothesized, and because in the pre-registration the hypothesis was non-directional, even though there a direction was insinuated by the other hypothesis? In general, is it best practice to make the hypothesis rejectable, or do you not want to reject it and just put the hypothesis in a way that it actually reflect hypothesized behavior?] Delegation shares around 50% also suggest that subjects, on average, find the equal relative performance information conveyed by the design credible and their behavior reflects neither systematic preference for their own predictions nor systematic deference to the algorithm.

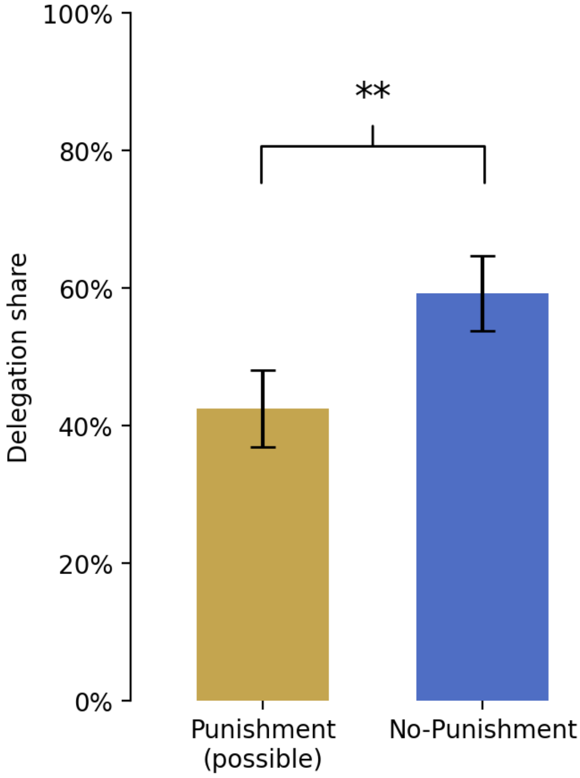
Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.*

With controls. Next, we examine the robustness of the observed treatment difference in the delegation decision. Specifically, we assess whether the difference in delegation rates remains when controlling for socio-demographic characteristics and Player A's actual and perceived task performance. Table 1 reports estimates from a Logit model of the delegation decision. Column (1) reports the unconditional treatment effect. Column (2) adds a vector of socio-demographic and technology-affinity controls. Columns (3) and (4) add Player A's actual performance (number of correct predictions in the first ten rounds) and her weighted-average belief about own performance, respectively. Column (5) restricts the sample to subjects who passed the attention check.[12] The treatment effect is robust across all five specifications: the coefficient on the No-Punishment indicator runs from 0.68 (p=0.034) in the unconditional specification

[11] A within-subject corroborator points the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment (25% of delegators) while only 5 would have started (15.2% of non-delegators), yielding a hypothetical delegation rate of 50.6% (within-subject McNemar's test $p = 0.090$). Given that this decision is purely hypothetical, the within-subject pattern complements rather than substitutes for the between-subject test.

[12] While the attention check was included on another screen, it may still correlate with who paid attention in deciding to delegate.

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the prediction for Player B to the algorithm in each condition. The two-sided difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher's exact test ($p = 0.041$).

to 0.83 (p=0.026) in the attention-check sample.[13] Translating these into average marginal effects: removing the punishment possibility raises the probability of delegation by between 16.4 and 19.1 percentage points across all specifications. Neither actual nor perceived performance is significantly associated with delegation in the pooled sample, although actual performance loads negatively and approaches significance under the attention restriction (p=0.074). We return to the heterogeneity-by-performance pattern in Section 4.1.

Table 1: Determinants of the delegation decision

	Dependent variable: *Delegation*				
	Full sample				*Passers*
	(1)	(2)	(3)	(4)	(5)
No-Punishment indicator	0.677**	0.696**	0.757**	0.721**	0.834**
	(0.320)	(0.329)	(0.334)	(0.333)	(0.376)
Task performance			−0.192		−0.254*
			(0.128)		(0.142)
Perceived performance				−0.118	
				(0.094)	
Age		−0.004	−0.004	−0.003	0.002
		(0.015)	(0.015)	(0.015)	(0.017)
Female		0.279	0.350	0.332	0.651
		(0.366)	(0.369)	(0.372)	(0.414)
Socio-economic status		−0.062	−0.058	−0.051	−0.134
		(0.113)	(0.116)	(0.115)	(0.130)
Went to uni		0.061	0.091	0.047	0.184
		(0.369)	(0.364)	(0.367)	(0.400)
Technology score		−0.110	−0.078	−0.101	0.021
		(0.149)	(0.153)	(0.148)	(0.174)
Leadership position		0.647*	0.676*	0.641*	0.413
		(0.364)	(0.367)	(0.365)	(0.412)
Constant	−0.302	0.036	0.402	0.449	0.241
	(0.226)	(0.862)	(0.930)	(0.945)	(1.062)
AME of No-Punishment (pp)	16.4**	16.5**	17.7**	16.9**	19.1**
	($p = 0.025$)	($p = 0.026$)	($p = 0.016$)	($p = 0.022$)	($p = 0.017$)
N	161	159	159	159	130
Pseudo R^2	0.020	0.039	0.049	0.047	0.063

Note: Coefficients from a Logit regression of the delegation decision on the No-Punishment indicator and the controls listed in each column. Robust (HC1) standard errors in parentheses. Significance: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$ (two-sided). Two subjects are dropped in Columns (2)–(4) due to missing socio-demographic data. The *AME of No-Punishment* row reports the average marginal effect of the No-Punishment indicator on the probability of delegation, in percentage points.

Player B – Punishment behavior

Next, we turn to Player B's punishment behavior in the *Punishment* condition, in which Player B was given a real, payoff-relevant punishment choice.

Punishment in all four (delegation, outcome) scenarios of the strategy method is significantly greater than zero (one-sample t-test $p < 10^{-5}$ in every scenario). Despite the perhaps surprisingly high average punishment for good outcomes, Player B punishes more for low-payoff

[13] Including a control for the random order in which the two delegation options appeared on the screen in leaves the treatment coefficient essentially unchanged (treatment coefficient 0.76 with p=0.022, with the order indicator at 0.43, p=0.22, starting from the specification in Column (3)).

outcomes than for high-payoff outcomes—with a within-subject mean difference of about £0.21, significant at the 1% level—an indication that punishment carries informational content rather than being random.[14] 45.0% of Player Bs do not punish in any of the four scenarios.[15]

Conditional on the outcome, however, Player A is not significantly differently punished depending on the delegation decision (Figure 3). When Player B receives the high payoff, average punishment is £0.329 for delegated decisions and £0.297 for self-made decisions. When Player B receives the low payoff, average punishment is £0.482 when Player A delegated versus £0.553 when she made the decision herself. This difference points in the direction of H2 but is not significant (paired t-test $p = 0.131$; Wilcoxon signed-rank $p = 0.544$). Table 2 reports an OLS regression of chosen punishment on a delegation indicator, an outcome indicator, their interaction, Player B's belief about Player A's performance, and a vector of socio-demographic controls (column 1). The interaction term is borderline statistically insignificant ($p = 0.077$), confirming the bivariate result. Even though the algorithm and Player A are calibrated to perform identically, Player Bs may still hold heterogeneous beliefs about Player A's *absolute* performance. We rule out the possibility that those beliefs drive cell-level punishment. Across all four scenarios, correlation between Player B's weighted average belief about Player A's score and chosen punishment are essentially zero ($|r| \leq 0.03$, all $p > 0.8$).

Equivalently, regressing the within-subject difference in punishment (self-made minus delegated) on a bad-outcome indicator yields a diff-in-diff estimate of £0.10, which is borderline insignificant at $p = 0.07$ (Table 2, column 2).[16] The same picture arises when measuring punishment frequencies rather than amounts (Figure 3, panel A, right axis) [Comment: Logit model in appendix?]. Thus, Player B neither punishes delegated decisions substantially less harshly, nor do they punish them less frequently.[17]

[Comment: Do we even need the regressions with the difference as the dependent variable?]

Therefore, in our experimental setting Player A cannot avoid punishment by delegating to the algorithm, nor does she seem to be punished more when delegating the decision to the algorithm. Instead, punishment is determined almost entirely by the realized outcome. This non-result also bears on *Result 1*: Any treatment difference in delegation frequencies motivated by punishment avoidance would not be consistent with actual punishment behavior of Player B.

Result 2. *Player A cannot avoid punishment by delegating the decision to the algorithm.*

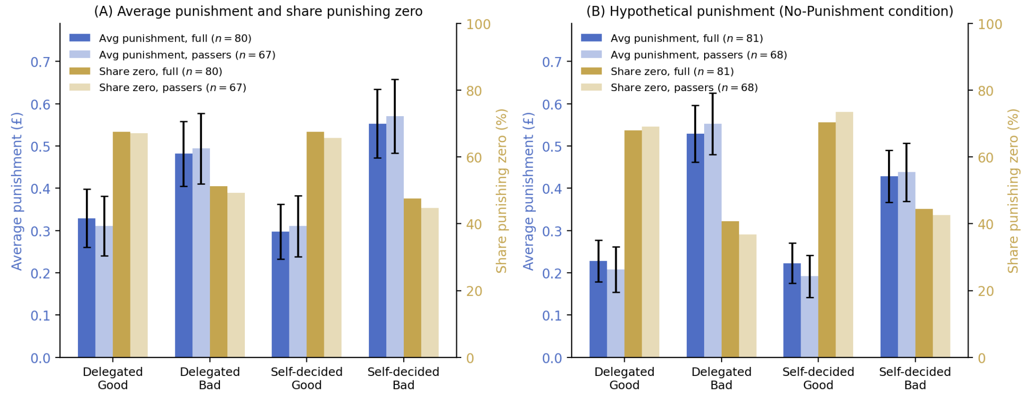
[14] Coffman (2011) similarly documents non-trivial punishment in scenarios where the decision-maker's action could not have been improved upon. Thus, positive punishment in the good-outcome cells need not reflect noise. Instead, Player Bs may hold ex-post views about whether the *delegation choice itself* was appropriate, independent of the realised outcome, and can punish accordingly.

[15] Among the 44 Player Bs who do punish in at least one scenario, the punishment levels are mostly non-trivial, with the costly punishment option therefore being used as a substantive sanction.

[16] Including socio-demographic controls and Player B's belief about Player A's performance leaves the diff-in-diff unchanged at +£0.10 ($p = 0.08$, column 3).

[17] As a robustness check, we re-estimate the analyses on the subset of subjects who passed the attention check on the punishment-elicitation screen. The result is essentially identical under this sample restriction (mean difference £0.077 vs £0.071; paired t-test $p = 0.153$ vs $p = 0.131$; Table 2, columns 5–6).

Figure 3: Player B punishment by delegation and outcome scenario



Note: Both panels report, for each of the four (delegation, outcome) cells of the strategy method, the average punishment chosen by Player B (left y-axis, in pounds, error bars are standard errors of the mean) and the share of Player Bs choosing zero punishment (right y-axis, in percent). In each panel, the full sample is shown at full opacity; the attention-check passers are overlaid at lower opacity. Panel A reports the actual punishment in the *Punishment* condition (full sample $n = 80$, passers $n = 67$). Panel B reports the hypothetical punishment in the *No-Punishment* condition (full sample $n = 81$, passers $n = 68$), where Player Bs answered the same questions but could not impose punishment in practice.

[Comment: What about comparing to hypothetical punishment decisions?]

4.1 Discussion of results

We not only fail to support *Hypothesis 1*, but the observed effect points in the opposite direction [Comment: This only holds for the one-sided hypothesis. If we now change to a two-sided hypothesis, we have to change the formulation here.], suggesting that the possibility of punishment discourages delegation to the algorithm. In this section we explore four [Comment: Is that true?] candidate mechanisms that... [Comment: complete, once the mechanism paragraphs are worked out.].

Departures from prior settings. Our hypothesis was grounded in previous empirical evidence, though these studies were conducted in settings that differ from ours in various important ways. [Comment: This only holds for the one-sided hypothesis. If we now change to a two-sided hypothesis, we have to change the formulation here.] Most notably, our task departs fundamentally from the modified dictator games used in Bartling and Fischbacher (2012); Coffman (2011); Hamman et al. (2010); Oexl and Grossman (2013). In such games, the decision intent is more transparent: participants make either fair or unfair allocations that are directly payoff-relevant to themselves and implemented with certainty. In contrast, our prediction task involves uncertainty (subjects may err even though they put in substantial effort) and the incentives of Player A and Player B are aligned.[18] These aspects may obfuscate the intent of Player A, possibly diluting an intention-based punishment motive for Player B, and thus the need to avoid punishment by Player A. Nevertheless, we chose a prediction task because prediction under

[18] We consider the incentives of Player A and B to be aligned, even though Player A's payoff is only indirectly affected by her own decision, insofar as a better outcome for Player B reduces expected punishment.

Table 2: Determinants of Player B's chosen punishment

	Dependent variable: *Punishment*					
	Full sample			*Passers*		
	Level (£)	*Difference* (£)		*Level* (£)	*Difference* (£)	
	(1)	(2)	(3)	(4)	(5)	(6)
Delegated	0.032			0.001		
	(0.052)			(0.054)		
Bad outcome	0.256***	0.103*	0.103*	0.260***	0.078	0.078
	(0.071)	(0.057)	(0.059)	(0.080)	(0.052)	(0.053)
Delegated × Bad outcome	−0.103*			−0.078		
	(0.058)			(0.052)		
Player B belief about Player A perf.	−0.020		−0.006	0.001		−0.006
	(0.032)		(0.029)	(0.037)		(0.038)
Age	0.002		−0.009**	−0.001		−0.009**
	(0.005)		(0.003)	(0.006)		(0.004)
Female	−0.100		−0.012	−0.064		0.003
	(0.128)		(0.109)	(0.140)		(0.132)
Socio-economic status	0.083*		0.029	0.072		0.041
	(0.047)		(0.024)	(0.049)		(0.028)
Went to uni	−0.196		0.051	−0.098		0.023
	(0.140)		(0.070)	(0.142)		(0.075)
Technology score	0.073		−0.022	0.030		−0.012
	(0.053)		(0.027)	(0.059)		(0.033)
Leadership position	0.083		0.143	0.061		0.174
	(0.142)		(0.122)	(0.164)		(0.145)
Constant	−0.077	−0.032	0.116	0.020	−0.001	0.076
	(0.328)	(0.051)	(0.320)	(0.374)	(0.053)	(0.397)
Observations	320	160	160	268	134	134
Subjects	80	80	80	67	67	67
Adj. R^2	0.077	0.008	0.038	0.030	0.001	0.043

Note: Columns (1) and (4): OLS regression of Player B's chosen punishment level on a delegation indicator, a bad-outcome indicator, their interaction, Player B's belief about Player A's performance, and a vector of socio-demographic controls; sample stacked over the four (delegation, outcome) cells of the strategy method (4 observations per Player B). Columns (2)/(5) and (3)/(6): OLS regression of the within-subject delegation insulation (*punish_nodel* − *punish_del*) on a bad-outcome indicator and, in Columns (3)/(6), additional controls; sample stacked over the two outcome cells (2 observations per Player B). In the difference specifications, the intercept is the average insulation in the good-outcome cell and the bad-outcome coefficient is the additional insulation in the bad-outcome cell. Columns (1)–(3) use the full Punishment-condition sample. Columns (4)–(6) restrict to Player Bs who passed the attention check on the punishment-elicitation screen, as preregistered. Standard errors clustered at the Player B level. Significance: $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$ (two-sided).

uncertainty is precisely the domain where algorithmic decision-making is most prevalent.

While this story is consistent with the obsrvation of a lack of a difference in punishment between delegated and self-made decisions, it can not explain the reversal, but merely the absence of any treatment effect in the likelihood to delegate. Moreover, it would be reflected in punishment beliefs, which we discuss next. [Comment: Given how noisy of a measure these beliefs are, I am not sure if we need beliefs to align with this story in order to believe it. Maybe take some weight away.]

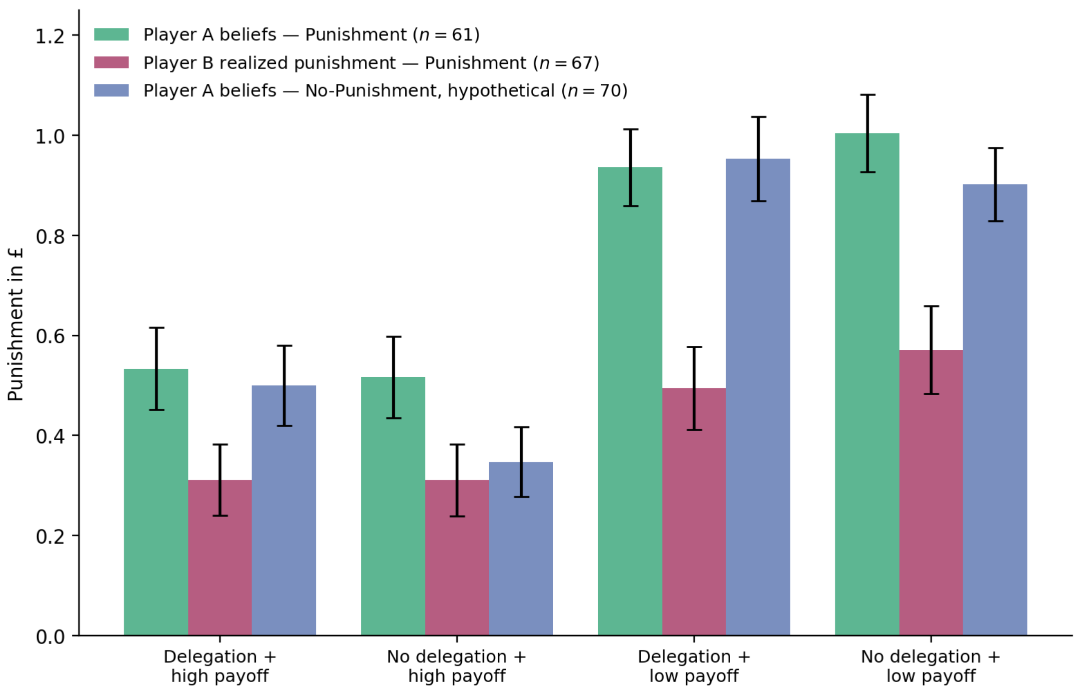
(M1) Anticipated differential punishment. A natural reading of Result 1 is that Player A anticipates stronger punishment for a delegated decision than for a self-made decision and delegates less in response. This belief pattern would be consistent with parts of prior work suggesting that observers view delegating moral decisions to algorithms relatively critically (e.g., Gogoll and Uhl, 2018; Niszczota and Kaszás, 2020; Jauernig et al., 2022) [Comment: check that this is indeed the case]. Alternatively, the same belief pattern would be consistent with a simple shirking motive, in which Player A expects greater punishment for shirking the effort to make a decision for Player B but, in the absence of punishment, can shirk without monetary consequences. We have already seen that Player B punishment is outcome-based, rather than delegation-based (which is not surprising given that performance is controlled for), but a shirking motive — which would predict extra punishment for delegation — could at least rationalize the positive punishment observed in the good outcome–delegated scenario. In either case, delegation behavior should be consistent with observed beliefs about punishment.

Punishment beliefs. After making the delegation decision but before learning the outcome, we elicited Player A's beliefs about Player B's punishment in each of the four scenarios.

Several caveats may temper the interpretation of the belief measure. The elicitation was un-incentivised and took place after the delegation decision, so reports may reflect hedging across scenarios or post-hoc rationalization rather than the beliefs that actually drove the choice. As a corollary, any within-subject correlation between beliefs and delegation cannot identify direction: we cannot distinguish whether subjects delegated *because* they held these specific beliefs about punishment, or hold these beliefs *because* they delegated. Hypotheticality compounds the noise: only two of the four *(delegation, outcome)* scenarios remained payoff-relevant after the delegation decision in the *Punishment* condition, and all four were purely hypothetical in the *No-Punishment* condition. We therefore restrict the belief–delegation analysis to the *Punishment* condition. The motivation is structural rather than statistical: the punishment-anticipation channel can only operate where punishment is possible, so beliefs about a counterfactual that cannot materialise do not enter Player A's decision rule in the *No-Punishment* arm. The exercise that follows should accordingly be read as suggestive evidence on whether the channel is alive among those exposed to it, not as a mediation analysis of the treatment-level delegation gap (Result 1).

Two patterns emerge. First, Player As substantially *overestimate* the level of punishment they will face: average expected punishment exceeds average realized punishment in every scenario (Figure 4, green vs. magenta bars). The same pattern is visible in the *No-Punishment* condition, where Player A's hypothetical beliefs (blue bars) closely track the Punishment-condition beliefs. Second, however, Player As correctly anticipate the *relative* pattern of punishment in that they expect no significant difference in punishment between delegated and non-delegated decisions, conditional on outcome—a pattern that closely tracks the actual punishment behavior of Player B documented in Result 2. *It is this relative-pattern finding—on punishment beliefs alone—that provide no support for a punishment anticipation mechanism for Result 1.* If Player A had reduced delegation in the *Punishment* condition because she anticipated being punished more harshly for delegation than for self-made decisions, we would expect to see this asymmetry in the elicited beliefs. We do not. [Comment: Let's consider to just leave out te hypotheical punishment beliefs. We do not do anything with them.]

Figure 4: Realised vs. anticipated punishment, by scenario



Note: Average punishment in pounds across the four *(delegation, outcome)* scenarios, attention-check passers only. Green bars: Player A's anticipated punishment in the *Punishment* condition ($n = 61$). Magenta bars: Player B's realised punishment in the *Punishment* condition ($n = 67$). Blue bars: Player A's hypothetical punishment beliefs in the *No-Punishment* condition ($n = 70$), shown for calibration only — these beliefs concern a counterfactual that cannot materialise and do not enter Player A's delegation decision in this arm. Whiskers denote ±1 standard error.

Even though the belief measure was not incentivized, and even though at the time of the belief elicitation, two of the four scenarios could not materialize anymore, punishment beliefs carry within-subject information. Table 3 reports a Logit regression of the delegation decision on Player A's belief difference (*no delegation − delegation*) in expected punishment, restricted throughout to the *Punishment* condition.[19] The first three columns use the simple average of

[19]The two underlying belief measures compare expected punishment of a self-made and delegated decision for

the two outcome-conditional belief differences as the single regressor; this maps cleanly onto a punishment-anticipation channel under expected-utility reasoning, in which only the expected punishment penalty for switching from delegation to self-decision matters. Column (1) reports the unconditional specification on the full *Punishment* sample; Column (2) adds task performance and socio-demographic controls; Column (3) restricts to subjects who passed the attention check on the belief-elicitation screen, as preregistered. Column (4) replaces the simple average with the more theoretically appropriate weighted average, using Player A's elicited probability of producing the high payoff as the weight (Pr(good) $= wa\_confidence/10$). Column (5) instead enters the two outcome-conditional belief differences separately.

[Comment: Maybe for consistency, include also here the marginal effect at the bottom, even thou now it would be for different variables.]

The pattern is consistent with the punishment-anticipation channel *conditional on attention*, but unconditionally weak. In Columns (1)–(2), the belief coefficient is positive but small and far from significant ($p > 0.39$): in the unrestricted *Punishment* sample, expected differential punishment does not predict delegation. Restricting to attention-check passers (Column (3)) sharpens the relationship substantially: the simple-average belief coefficient is 2.71 ($p = 0.035$), implying that subjects who expect a one-pound larger punishment penalty for not delegating are markedly more likely to delegate. The weighted-difference specification (Column (4)) yields a similar conclusion with a slightly larger coefficient (3.33, $p = 0.030$), supporting the expected-utility interpretation. In Column (5), the good-outcome belief difference carries the regression: its coefficient of 1.58 ($p = 0.033$) is in line with the single-regressor specifications, while the bad-outcome belief difference is positive but statistically noisier (0.94, $p = 0.21$). If taken at face value this would mean that the delegation decision is primarily made by thinking about punishment consequences after a good outcome, rather than a bad outcome.

Interpretation of the two-difference decomposition. The five-column table treats the simple- and weighted-average specifications as the primary structural mappings of the channel and the two-difference decomposition (Column (5)) as a descriptive complement. Two interpretive points are worth flagging. First, the two outcome-conditional differences are nearly orthogonal in this sample (Pearson correlation -0.10 in the passers sample), so the two coefficients in Column (5) are essentially additive partial slopes rather than competing for the same variance. Second, under expected-utility reasoning only the weighted sum of the two differences should enter the decision rule; entering both separately therefore answers the partial question "conditional on the bad-outcome asymmetry, how does delegation respond to the good-outcome asymmetry?" rather than the structural question the channel poses. We therefore lead with the simple-average and weighted-average specifications and treat the decomposition as a robustness check that confirms the action sits in the good-outcome cell. [Comment: this whole pararaph should become a footnote.]

---

each of the two possible outcomes for Player B, so that higher values correspond to higher expected punishment for non-delegated decisions relative to delegated decisions.

Table 3: Delegation and punishment beliefs

	Dependent variable: *Delegation*				
	Full Punishment		*Punishment, passers*		
	(1)	(2)	(3)	(4)	(5)
Punishment beliefs: simple average ($\overline{\Delta}$)	0.553	0.629	2.708**		
	(0.656)	(0.744)	(1.286)		
Punishment beliefs: weighted by Pr(good)				3.329**	
				(1.532)	
Punishment beliefs: bad outcome (no del. − del.)					0.937
					(0.751)
Punishment beliefs: good outcome (no del. − del.)					1.579**
					(0.741)
Task performance		−0.337*	−0.576**	−0.544**	−0.609***
		(0.181)	(0.231)	(0.245)	(0.236)
Age		0.037*	0.063**	0.066**	0.064**
		(0.020)	(0.029)	(0.030)	(0.030)
Female		0.213	1.522**	1.630**	1.334*
		(0.592)	(0.754)	(0.699)	(0.793)
Socio-economic status		−0.079	−0.012	−0.095	−0.003
		(0.185)	(0.226)	(0.233)	(0.219)
Went to uni		0.108	−0.446	−0.257	−0.433
		(0.609)	(0.712)	(0.719)	(0.707)
Technology score		−0.104	0.085	0.129	0.059
		(0.230)	(0.290)	(0.295)	(0.283)
Leadership position		0.478	−0.485	−0.477	−0.387
		(0.510)	(0.720)	(0.731)	(0.729)
Constant	−0.316	−0.536	−2.023	−2.153	−1.842
	(0.228)	(1.333)	(2.080)	(2.052)	(2.103)
Controls	No	Yes	Yes	Yes	Yes
Sample	*Punishment*	*Punishment*	Passers	Passers	Passers
N	80	78	60	60	60
Pseudo R^2	0.006	0.097	0.241	0.258	0.246

Note: Logit estimates with robust (HC1) standard errors in parentheses. Significance: $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$ (two-sided). Sample restricted to the *Punishment* condition throughout, since beliefs about a counterfactual that cannot materialise do not enter Player A's decision rule in the *No-Punishment* arm. Belief differences are within-subject *(no delegation)* − *(delegation)* in expected punishment, so positive coefficients indicate that subjects who expect to be punished more harshly for not delegating (relative to delegating) are more likely to delegate. Column (1) uses the simple average of the two outcome-conditional belief differences without controls; Column (2) adds task performance and socio-demographic controls; Column (3) restricts to subjects who passed the attention check on the belief-elicitation screen, as preregistered; Column (4) replaces the simple average with a weighted average using Player A's elicited probability of producing the high payoff (Pr(good) = *wa_confidence*/10); Column (5) replaces the single belief regressor with the two outcome-conditional belief differences entered separately.

Taken together, the belief evidence is mixed: within the Punishment sample, individual heterogeneity in belief differences does predict delegation in the direction M1 predicts (Table 3, Cols. 3–4), but this within-condition signal is not conclusive for the treatment-level gap (Result 1). Moreover, Player B does not deliver the asymmetric punishment that would justify the belief asymmetry on which the channel rests (Result 2). Anticipated differential punishment as the driver of Result 1 is therefore only partly consistent with elicited beliefs and inconsistent with realized punishment.

Heterogeneity by performance. The regression framework also lets us explore which other factors influence the delegation decision. Already in Table 1 we observe a weak negative correlation between delegation and Player A's task performance across conditions, which is borderline significant when restricting the sample to subjects passing the attention check. Within the *Punishment* condition, this relationship is even more pronounced and Player As who performed better on the first ten rounds were less likely to delegate.[20] In the *No-Punishment* condition, the relationship between performance and delegation is much less pronounced. Figure 5 visualises this asymmetry.

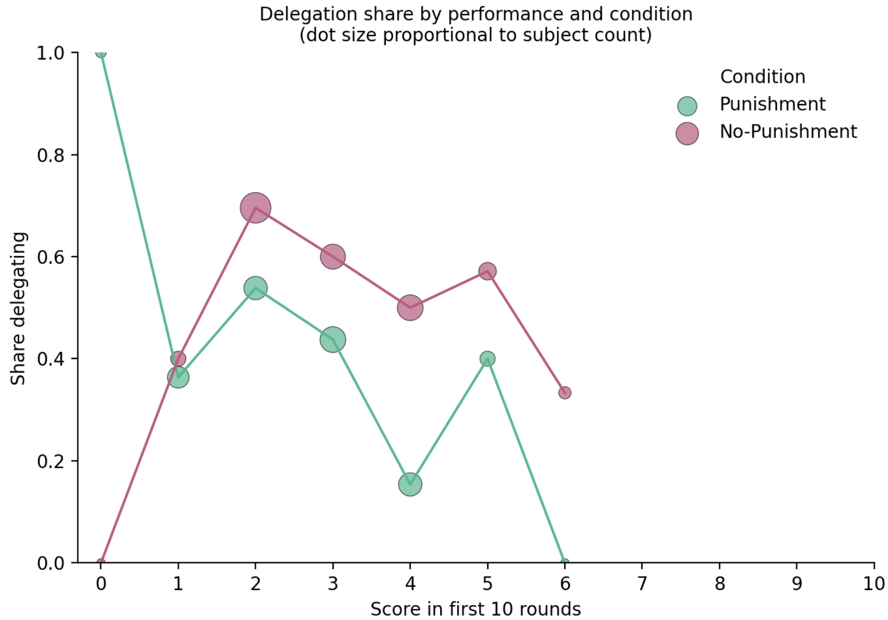
This contrasts with the setting of Feier et al. (2022), in which the algorithm's performance distribution is matched to the human distribution at the *population* level: a high-performing subject who recognizes her position in that distribution rationally believes she outperforms the algorithm in expectation and a low-performing one rationally believes the reverse, producing an expected-utility-rational performance-to-delegation gradient. In our design, performance is equalised at the *individual* level and is common knowledge, so expected utility predicts no gradient at all; the pattern documented here in the *Punishment* condition is therefore informative about preferences rather than about beliefs over relative ability.

(M2) Process ownership of the outcome. This pattern suggests that Player As who have a higher chance of producing the high payoff for Player B prefer to bring about that good outcome themselves rather than via the algorithm. The intuition draws on the psychological-ownership literature: people derive utility from being the proximate cause of a good outcome for another, and this utility is asymmetric with respect to outcomes (Steffel et al., 2016). The pattern is consistent with process ownership being behaviourally consequential only when Player B's payoff is at meaningful third-party stake — present in the *Punishment* arm, where Player B can sanction; absent in the *No-Punishment* arm, where Player B receives the payoff but has no enforcement role. Importantly, this does not require Player A to anticipate *differential* punishment for delegated vs. self-made decisions; the elicited beliefs show no such asymmetry. [Comment: check Steffel paper: Steffel (2016) is cited, but Steffel's setting is about delegating decisions through agents — whether her finding specifically supports "stake-conditional acti-

[20] A similar relationship holds for *perceived* performance: Player A's weighted-average belief about her own task performance is negatively related to the delegation decision, e.g. coef. = -0.06 in the passers with controls specification

vation" of process ownership is worth verifying. If it doesn't, weaken to "consistent with the broader psychological-ownership literature, in which intensity of caring tracks the size of the stake.]

Figure 5: Delegation share by performance and condition



Note: Each point reports the share of Player As at the corresponding first-ten-rounds score who delegated the final prediction. Dot size is proportional to the number of subjects at that score. The downward slope in the *Punishment* condition is the Pearson $r = -0.19$ ($p = 0.092$); no comparable slope in *No-Punishment*. Restricted to attention-check passers.

Finding this relationship in the *Punishment* condition only indicates that Player A does not merely avoid directly causing a bad outcome and savoring directly causing a good outcome for Player B. Instead, it suggests a connection to anticipated punishment, such that either (i) Player A expects more severe punishment for delegated decisions (relative to non-delegated ones) when the outcome is good than when it is bad, or (ii) better performing Player As hold systematically different punishment beliefs than worse-performing Player As. However, neither do we find evidence that Player As expect a relatively larger punishment penalty for delegation in good-outcome states than in bad-outcome states, nor do Player A's punishment beliefs do vary systematically with Player A's performance.[21]

[Comment: Think more about this potential non-linearity, i.e. If I know it will fail, then I may want to delegate. If I know I will succeed, I may want to do it myself. Hence, my preference on delegation may change depending on the chance of success. What kind of literature is here? This will also relate to the point about the amount of inherent uncertainty in the decision.]

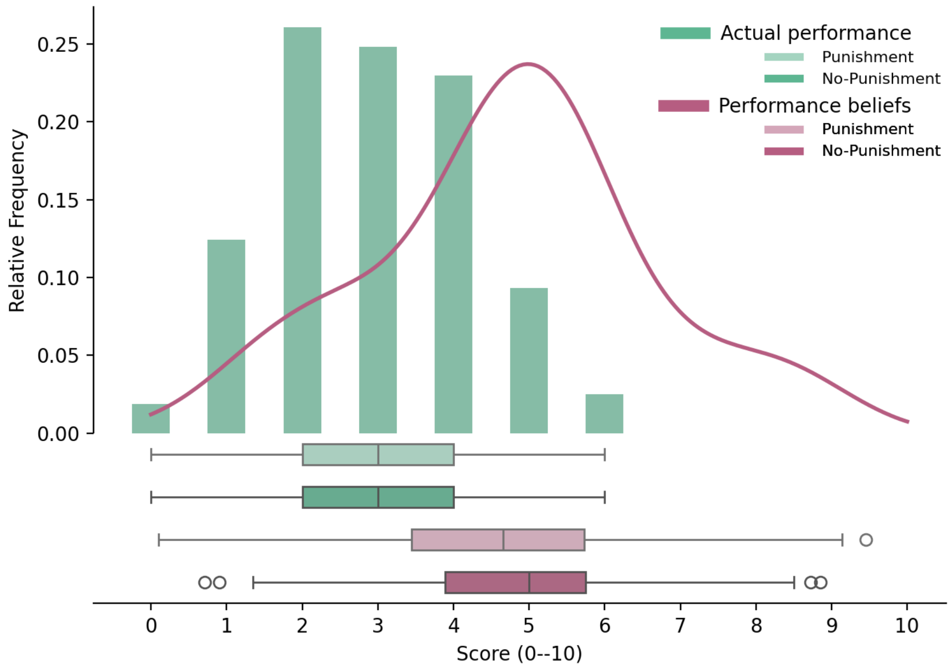
Inherent uncertainty. Alternatively, differences in the chances of success for Player B may also be interpreted as differences in uncertainty inherent in the task. Dietvorst and Bharti

[21] See Appendix A.6 for the supporting numerical and graphical evidence: a within-subject test that $\Delta^{\text{good}} = \Delta^{\text{bad}}$ ($p = 0.376$ in the passers sample), and all six correlations between Player A's punishment beliefs and her task performance ($|r| \leq 0.21$, none significant at conventional levels).

(2020) provide suggestive experimental evidence on how the amount of uncertainty inherent in a decision affects the preference for algorithms. They find that with increasing task uncertainty, subjects' preference for algorithms over their own prediction decreases, even though they were aware that the algorithm performs better on average. In our setting, arguably, uncertainty is maximized at a score of 50%, and most Player A's perform worse, so that the performance distribution largely sits below 50%. In turn, better performance comes with greater outcome uncertainty and is linked to lower delegation rates. Therefore, the mechanism Dietvorst and Bharti (2020) describe is consistent with what we find in a setting in which the decision is payoff-relevant to another participant, equal relative performance is common knowledge and punishment is possible. However, this reasoning would apply to treatment and baseline equally and thus not explain Result 1.

(M3) Performance differences and overconfidence across conditions. If delegation is to some extent dictated by task performance, Result 1 could be a direct artifact of performance differences between treatments. However, treatment differences in delegation are not driven by performance differences across conditions in the first ten rounds. The average subject scores correctly in roughly three of the first ten rounds and performance is balanced across conditions (Figure 6). Subjects also exhibit significant overconfidence, measured by the weighted-average belief about own performance; while this overconfidence may partly reflect the information that the algorithm is performing equally well, it is not statistically different across conditions.

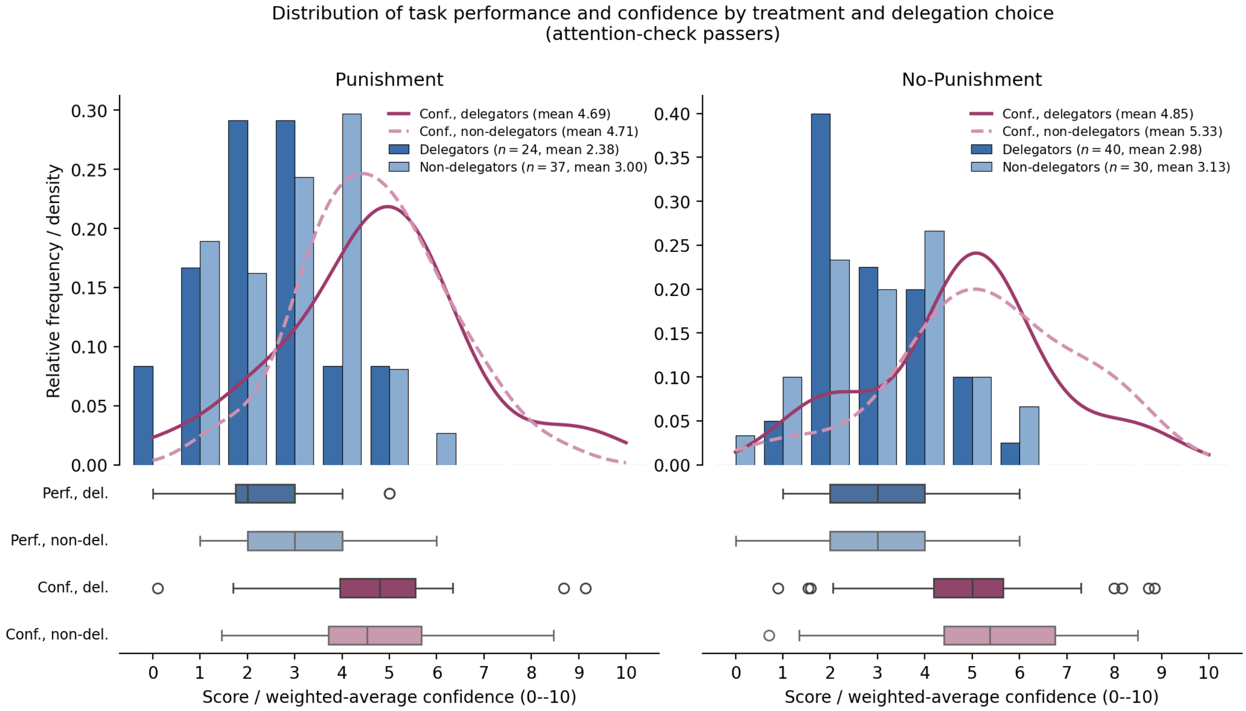
Figure 6: Performance in the first ten rounds, by treatment



Note: Distribution of correct predictions (within 10 lbs of the true weight) over the first ten task rounds, by condition. Average performance is 2.93 correct out of ten and is statistically indistinguishable across conditions.

[Comment: Alternative figure]

Figure 7: Performance and confidence distributions by treatment and delegation choice



Note: Restricted to attention-check passers. Within each treatment, the top panel shows the share of Player As at each first-ten-rounds score, separately for those who delegated (dark blue) and those who did not (light blue), with the weighted-average performance belief overlaid as a KDE (delegators solid magenta, non-delegators dashed pink). Boxplot strips below the top panel report the same four series. In the *Punishment* condition, actual performance discriminates between delegators and non-delegators (means 2.38 vs. 3.00) while perceived performance does not (means 4.69 vs. 4.71). In the *No-Punishment* condition, neither actual nor perceived performance discriminates (actual: 2.98 vs. 3.13; perceived: 4.85 vs. 5.33). The additional delegators in *No-Punishment* therefore come disproportionately from the middle and upper portion of the actual performance distribution.

(M4) Effort exertion to outperform the algorithm. Although our design controls for relative performance based on the first ten rounds—making the algorithm and Player A equivalent in expectation on the final prediction—potential punishment may motivate Player A to exert greater effort in making the decision for Player B, thereby choosing not to delegate in hopes of outperforming the algorithm.

If effort, measured by performance, increased in both conditions, effort exertion could be interpreted as altruistic, and would be treatment-independent. However, if effort only increases with punishment, then the motive would be driven by the expectation to decrease the likelihood of punishment, which could explain Result 1.

The data are inconsistent with the simple form of this story. Measuring performance change as the difference between the average absolute prediction error across the first ten rounds and the absolute prediction error on the final prediction, we find that Player As in the *No-Punishment* condition improve their performance from rounds 1–10 to round 11, whereas Player As in the *Punishment* condition do not improve. Because all Player As across both conditions faced the same image on the final prediction, this differential improvement cannot be attributed to task-specific variation. Therefore, increased effort appears concentrated among non-delegators in the *No-Punishment* condition, where punishment is *not* on the table—incompatible with a story in which Player A in the *Punishment* condition tries especially hard to outperform the algorithm in order to avoid punishment. In fact, among Player As who chose not to delegate, those in the *No-Punishment* condition (where punishment is not possible and the delegation share is higher) outperformed those in the *Punishment* condition: 45.5% of non-delegating Player As in the *No-Punishment* condition earned the high payoff for Player B, compared to 21.7% in the *Punishment* condition. This difference is not driven by ex-ante performance differences across the first ten rounds of those that decided not to delegate (*No-Punishment*: 3.12/10; *Punishment*: 2.98/10; two-sided $p = 0.66$).[22]

Thus, treatment differences in the propensity to delegate do neither stem from ex-ante differences in performance between both treatments, nor from increased effort for the decision for Player B in the *Punishment* condition.

In sum, we considered four candidate mechanisms behind Result 1. M1 (anticipated differential punishment) is inconsistent with Player A's elicited beliefs: she does not expect a relatively larger punishment penalty for delegation in the good-outcome state than in the bad-outcome state. M2 (process ownership of the outcome) is the only candidate the data support: the negative performance–delegation gradient in *Punishment* and the absence of any comparable gradient in *No-Punishment* (Figure 5, Figure 7) jointly indicate that Player As who are likely to deliver the high payoff to Player B prefer to do so personally, but only when Player B's

[22]Player As who chose to delegate in the *Punishment* condition scored on average 2.47/10 in the first ten rounds, compared with 3.06/10 for delegators in the *No-Punishment* condition (two-sided p=0.039, p=0.080 in passers). The extra delegators in *No-Punishment* therefore come disproportionately from the middle and upper portion of the performance distribution — subjects who, when punishment is on the table, choose to make the decision themselves.

payoff is at meaningful third-party stake. M3 (ex-ante differences in task performance between conditions) is inconsistent with the balance documented in Figure 6: average performance over rounds 1–10 is statistically indistinguishable across treatments. M4 (extra effort to outperform the algorithm under the threat of punishment) is inconsistent with the round-11 effort data, which run in the opposite direction: it is in the *No-Punishment* condition, not in the *Punishment* condition, that non-delegators improve from rounds 1–10 to the final prediction. However, we are explicit that the supporting evidence for M2 consists of a single heterogeneity pattern in a sub-sample and a comparison of delegator success rates; the pattern is the most consistent reading of the data we have, but not a confirmed mechanism.

Two channels of responsibility-shifting. Recent work suggests that what the prior literature has called "the responsibility-shifting motive" is in fact two distinct sub-motives that tend to co-move when the delegate is human but can decouple when the delegate is an algorithm. Steffel et al. (2016) document that delegation reduces both *felt* responsibility (the principal's own anticipated moral burden) and *anticipated blame* (the principal's anticipated third-party judgement), and that both mediators contribute independently to the propensity to delegate. Hüholt and Szech (2026) test the first channel in a moral-domain delegation paradigm and find that subjects delegate to AI *more* often when the moral consequences of the decision become real (rather than hypothetical), and rate AI's moral capability higher in real-stakes conditions — consistent with belief adaptation that lets the principal shed her *felt* moral burden through delegation. Our results speak to the second channel: when third-party punishment is possible, delegation to the algorithm *falls* (Result 1), and Player B punishes by outcome rather than by delegation choice (Result 2). Both are consistent with principals correctly anticipating that the algorithm does not effectively absorb *external* blame. Taken together, this suggests a two-channel reading in which AI is a strong shield against the principal's internal moral burden but a weak shield against the external accountability she faces from third parties — channels conflated in canonical human-delegate settings but separable when the intermediary is algorithmic.

4.2 Other

Order effects. The delegation decision and the punishment / belief-elicitation screens presented their options in randomised (or block-wise randomised) order. Order effects are localised to the *Punishment* condition: presenting the algorithm option first roughly doubles delegation (52.5% vs 32.5% when the own-decision option is first; χ^2 $p = 0.070$), while the *No-Punishment* condition shows no order effect ($p = 0.674$); the treatment $\times$ order interaction is not significant ($p = 0.115$), so the H1 finding is not driven by interaction with order.

Power.

Add an ex-post (or, preferably, the original ex-ante) power analysis here, framed as in Feier et al. (2022).

[To resolve:]

- For two subjects, we must be missing realizations of control variables or something, because the sample size decreases from 161 to 159. Do we leave this unaddressed or should we include a footnote?

- Inequality/deservingness motive: Player A may think that Player B does not have to work to potentially earn 5 pounds and consider this unfair. However, Player A does not know what else Player B did have to do. Player B may want to bring down Player A's payoff so that they earn similarly, but also here there is uncertainty and this would just be outcome-based and not prescribe a treatment effect [Comment: Do not discuss this.]

- Is there any benefit in looking at hypothetical punishment of the No-Punishment treatment? Given that there are a lot of hypotheticals also in the Punishment treatment, should we perhaps not even pool for any analysis that only looks at anticipated punishment, however not when relating the delegation decision to anticipated punishment [Comment: Decause if it is entirely hypothetical, why would the beliefs matter for the delegation decision?]; How can we exploit that at the belief elicitation stage, a subset of scenarios are already for sure not relevant, e.g. we could check whether the difference between good and bad for delegated decisions depends on whether the decision was delegated, and the same for non-delegated decisions.

- Similarly to Player A, Player B overestimates Player A's task performance. Therefore, the likelihood with which the good outcome scenario marterializes is anticipated to be higher than it actually is. Can we do anything with this? [Comment: If Player B thinks the task is easy, they could punish more for bad outcomes. Punishment (all four individually, but also differences) is independent (Pearson and Kendall) of beliefs about Player A's performance, so this holds consistently across the whole (mostly supported between 2 and 6) distribution of performance beliefs. Hence, from a punishment perspective it doesn't matter whether you delegate when task is easy or harder. For those thinking longer about the punishment decision, there is a positive correlation between easiness of the task and punishment for a bad outcome when you don't delegate vs. when you delegate.] Of course, the perception is again tainted by Player B being told that performance was the same.

- What can we do with the hypothetical delegation choice of subjects in the No-Punishment treatment? Empirically, the result moves into the same direction (ttest p-value: 0.089)]

- Are risk attitudes related to any decisions? More risk-seeking leading to less delegation maybe or what could be the hypothesis? [Comment: Could be interpreted as a coefficient as part of the main result 1 regression.] More risk-seeking people could also risk being punished more, but I do not see how that would relate to the results. [Comment: Result in Feier et al.: risk averse subjects were also more likely to delegate.]

- Do robustness analysis with performance not in points, but in terms of distance from the truth.

- There are hedging motives in the performance elicitation task, but let's ignore those.

- Ways to filter: (i) failed attention checks, (ii) did not spend a lot of time on the page (delegation page and beliefs about punishment/punishment page)

- Player B: Is punishment correlated with any socio-demographics. Shoud we not do a regression here as well? Related to this: For those that state that individuals should be held responsible for AI-generated decisions responsibility avoidance is less possible (pos. correlation between 'nodel-del-bad' and 'responsibility') (maybe should recode variable 'responsibility').

- Main result 1 is consistent with Gogoll and Uhl 2018: Algorithm use was punsihed more.

- Check: In python, use `ttest rel` instead of `ttest ind` to compare punishment.

- Check: Feier et al. use GLM Logit IRLS. Is that different from what I am using?

- Maybe check if Player As that are better/more confident in last guess have different beliefs.

- Scrutinize Feier et al. analysis, e.g. They have the same roles: Is punishment behavior independent of delegation behavior? Is it in line with delegation behavior?

- When showing balance across observables, then be careful to use chi-squared test for categorical variables.

- Show that randomization also worked on time taken, etc?

- For each analysis, make explicit whether this is with the reduced sample (attention checks) or full sample. Always consider also presenting the robustness check with the other.

[Design issue: Why not have punsihment beliefs before delegation decision?] [Design question: Would it have been interesting to compare the delegation behavior between a decision for oneself vs. someone else?]

[Consideration for literature review: Regarding Feier et al: If I always say that punishment in their experiment could be due to the signal value in delegation (concerns regarding relative performance), then this holds for both their treatments so should technically not explain why they find differential rewarding in the machine treatment.]

5 Discussion and Conclusion

This paper asks two questions about how the prospect of punishment interacts with the use of algorithms as decision aids. Can decision-makers shift responsibility for bad outcomes onto algorithms by delegating their decisions? And does the anticipation of being held responsible motivate delegation in the first place? We address both questions in a preregistered online experiment in which Player A makes a payoff-relevant prediction for Player B and may delegate the prediction to an algorithm calibrated to perform as well as Player A herself. A between-subject manipulation switches off punishment by Player B in the *No-Punishment* condition; the *Punishment* preserves the natural setting in which the prospect of punishment is on the table.

Two findings emerge. First, and most striking, the *prospect* of punishment *reduces*, rather than increases, the propensity to delegate to the algorithm: 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not. This direction is the opposite of what the responsibility-avoidance literature would predict, and we view the reversal as the paper's main empirical contribution—noting that, because the result reverses the preregistered direction, the analyses we use to interpret it (Section 4.1) are exploratory rather than confirmatory. Second, conditional on the outcome for Player B, we cannot reject the hypothesis that delegated and self-made decisions are punished equally. The point estimate runs in the predicted direction (lower punishment under delegation in the bad-outcome cell) but is small relative to the test's confidence interval at the achieved sample size; we therefore frame Result 2 as a failure to detect insulation rather than as positive evidence that delegation cannot insulate. Player A does not appear to be able to use this particular algorithm as a meaningful blame shield in the bounded sense the test can assess.

Mechanism. Why does punishment reduce, rather than increase, delegation? The data are inconsistent with a simple anticipation mechanism: Player As do not on average expect harsher punishment for delegated decisions, and they correctly anticipate the absence of differential punishment that we observe in Player B's actual choices. The data are likewise inconsistent with the simple form of an effort story in which the prospect of punishment motivates Player A to outperform the algorithm: among non-delegators, those in the *No-Punishment* condition (where punishment is impossible) outperform those in the *Punishment* condition, the reverse of what such a story implies. The reading most consistent with the data is one of *process ownership*: Player As who actively choose not to delegate take responsibility for the outcome, exert correspondingly higher effort, and that share is higher when punishment is on the table. The supporting evidence is exploratory rather than confirmatory, however, and rests on a single performance-by-delegation heterogeneity pattern in a 60-subject sub-sample (Column (3) of Table 1); we cannot, with this design, separate process ownership cleanly from an experimenter-demand reading in which the punishment cue itself nudges Player A toward direct engagement. We therefore present process ownership as the most consistent reading of the data we have, not as a confirmed mechanism. Disentangling it from demand is a clean direction for follow-up work, sketched below.

This refinement is consistent with, but goes beyond, the standard reading of the responsibility-avoidance literature. The classic finding—that delegation reduces punishment of the principal and that this prospect motivates delegation—was established in dictator-game settings where the underlying choice is between a known fair and a known unfair allocation, the intermediary is human, and the principal can be held accountable specifically for choosing a particular intermediary (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013). When the intermediary is an algorithm calibrated to mimic the principal and the underlying decision involves genuine uncertainty, none of the standard mechanisms through which delegation can shift responsibility—signal value of the delegation choice, the human-human punishment channel, the certainty of the unfair outcome—are available. What remains is a baseline outcome-based punishment, supplemented by an asymmetric desire to be the proximate cause of good outcomes when the moral stakes are salient.

Implications, with explicit bounds. Our findings speak to the design of accountability regimes governing algorithmic decision-making. The European Union's AI Act (European Union, 2024) and analogous regulatory frameworks elsewhere place increasing emphasis on assigning responsibility to human users of high-risk algorithmic systems. The implicit theory of behavior underlying these provisions is that holding humans accountable for algorithm-mediated decisions *disciplines* their use of those algorithms. Our results offer a behavioral complement to this view: when punishment is on the table, decision-makers in our setting do not simply delegate-and-blame the algorithm; they choose to take direct responsibility for the decision more often. From the perspective of efficiency, this need not be welfare-increasing: in domains where algorithms genuinely outperform humans, holding humans liable for outcomes may push decision-makers away from the more accurate technology. The trade-off between encouraging algorithmic adoption and assigning meaningful responsibility is, on this reading, sharper than the existing literature has suggested.

We note explicitly the bounds of this reading. Our evidence is a single online experiment with 322 UK Prolific participants on a stylised weight-prediction task, with stakes of £5 for Player B and a punishment range of £0–£2. The high-stakes algorithmic domains that motivate the regulatory debate—medical diagnosis, judicial risk assessment, hiring, credit allocation—differ from our setting in stake magnitude, professional context, expert training, the institutional structure of accountability, and the kind of population at the decision margin. We therefore view the result as identifying a behavioral channel that exists in principle and is detectable in our setting, not as a calibration of how big a lever accountability is in any specific applied domain. Mapping the magnitude of the effect against external benchmarks—e.g., physician adoption of diagnostic algorithms under malpractice exposure, or loan-officer deviation from credit-scoring algorithms under fair-lending audit—would require either domain-specific replication or instruments to which we do not have access. We flag this gap explicitly so that readers can calibrate the policy implication appropriately.

Limitations. Several caveats are in order. The belief measure we elicit is unincentivised and noisy; it conveys ordinal patterns reliably but cannot support fine-grained quantitative claims about Player A's anticipations. Two of the four belief scenarios are hypothetical conditional on the delegation decision, and all four are hypothetical in the *No-Punishment* condition; this further limits the inference, and we plan to report the H2 patterns separately for the payoff-relevant and the hypothetical cells once the analysis pipeline is in place. We use the strategy method for Player B's punishment decision, which is standard in this literature but may yield different absolute levels of punishment than a direct elicitation. Player A also makes the final prediction whether or not she ultimately decides to delegate, which equates the time cost of the two options but may signal her commitment differently than a design in which the prediction is skipped under delegation. We did not include a separate manipulation-comprehension check verifying that Player A internalised the punishment-on/off distinction; we relied on the prominent screen-level framing and on attention-check pass rates as indirect evidence, but a future replication should add this check. The absolute stakes are modest (£5 high payoff, £0–2 punishment range): we cannot speak to how the effect scales with stakes, and a stake-magnitude robustness check is the first item in the future-research list below. Our sample is drawn from UK Prolific participants, which limits external validity along the dimensions in which Prolific samples differ from broader populations and from professional decision-makers in the high-stakes algorithmic domains we discuss above.

A natural concern is whether the punishment possibility cues Player A to behave more conservatively for reasons unrelated to the substantive question—an experimenter-demand effect. We cannot rule this concern out from the data. The direction of any such effect is also ambiguous: a decision-maker who is reminded of being judged could plausibly delegate *more* (to shed responsibility) or *less* (to demonstrate effort and commitment); the literature does not unambiguously favor one over the other. We note two patterns weakly inconsistent with a strong demand reading—behavior does not track elicited beliefs as tightly as a pure-demand story would predict, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) is harder to motivate from demand alone—but we treat these as suggestive rather than dispositive. A direct test in the spirit of de Quidt et al. (2018) would be the natural next step.

Directions for future research. Several extensions would strengthen the inference. A direct-method companion would test the robustness of Result 2 against the strategy-method elicitation, and a meaningfully larger sample would let us put a tight confidence interval on the small but persistent point estimate of insulation under delegation. A condition in which Player A is told explicitly that Player B is unaware of the equal-performance calibration would test whether the relative-performance signal is the binding margin distinguishing our results from Feier et al. (2022). Conditions that experimentally vary the algorithm's performance relative to Player A (above, equal, below) would map the generality of the process-ownership reading. A demand-effect probe along the lines of de Quidt et al. (2018) would put a credible upper bound on the contribution of M4. A stake-magnitude robustness check, varying the £5/£1 high/low payoffs and the £0–£2 punishment range, would speak to whether the effect scales with stakes in the direction that policy-relevant settings would imply. Finally, replication in a different country, in a different decision domain (e.g., a medical-triage or hiring vignette), or with a sample of professional decision-makers rather than online crowd-workers would speak directly to the external-validity bounds we describe above.

Taken together, our results suggest a more textured story than the one that has emerged from the existing literature on responsibility avoidance and algorithm aversion. Within the bounds of the setting we study, the capacity to use algorithms as a tool for shifting responsibility appears limited, and the prospect of being held responsible—rather than encouraging delegation as a hedge—appears to push decision-makers towards direct engagement with the consequential choice. Whether the same channel operates in the high-stakes algorithmic domains that animate the regulatory debate is an empirical question that this study can frame but cannot, on its own, answer.

References

Bartling, B. and U. Fischbacher, "Shifting the Blame: On Delegation and Responsibility," *The Review of Economic Studies*, January 2012, *79* (1), 67–87.

Bartling, Björn, Urs Fischbacher, and Simeon Schudy, "Pivotality and responsibility attribution in sequential voting," *Journal of Public Economics*, 2015, *128*, 133–139.

Berger, Benedikt, Martin Adam, Alexander Rühr, and Alexander Benlian, "Watch Me Improve — Algorithm Aversion and Demonstrating the Ability to Learn," *Business & Information Systems Engineering*, 2021, *63* (1), 55–68.

Bigman, Yochanan E. and Kurt Gray, "People Are Averse to Machines Making Moral Decisions," *Cognition*, 2018, *181*, 21–34.

Bockstedt, Jesse C. and Joseph R. Buckman, "Humans' Use of AI Assistance: The Effect of Loss Aversion on Willingness to Delegate Decisions," *Management Science*, 2025, *72* (1), 323–342.

Bogert, Eric, Aaron Schecter, and Richard T. Watson, "Humans Rely More on Algorithms than Social Influence as a Task Becomes More Difficult," *Scientific Reports*, 2021, *11*, 8028.

Burton, Jason W., Mari-Klara Stein, and Tina Blegind Jensen, "A systematic review of algorithm aversion in augmented decision making," *Journal of Behavioral Decision Making*, April 2020, *33* (2), 220–239.

Castelo, Noah, Maarten W. Bos, and Donald R. Lehmann, "Task-Dependent Algorithm Aversion," *Journal of Marketing Research*, 2019, *56* (5), 809–825.

Chen, Daniel L., Martin Schonger, and Chris Wickens, "oTree — An Open-Source Platform for Laboratory, Online, and Field Experiments," *Journal of Behavioral and Experimental Finance*, 2016, *9*, 88–97.

Chevrier, Mathieu and Vincent Teixeira, "Algorithm Delegation and Responsibility: Shifting Blame to the Programmer?," 2024. GREDEG Working Paper No. 2024-04, Université Côte d'Azur.

Chugunova, Marina and Daniela Sele, "We and It: An interdisciplinary review of the experimental evidence on how humans interact with machines," *Journal of Behavioral and Experimental Economics*, August 2022, *99*, 101897.

Coffman, Lucas C, "Intermediation Reduces Punishment (and Reward)," *American Economic Journal: Microeconomics*, November 2011, *3* (4), 77–106.

de Quidt, Jonathan, Johannes Haushofer, and Christopher Roth, "Measuring and Bounding Experimenter Demand," *American Economic Review*, 2018, *108* (11), 3266–3302.

Dietvorst, Berkeley J. and Soaham Bharti, "People Reject Algorithms in Uncertain Decision Domains Because They Have Diminishing Sensitivity to Forecasting Error," *Psychological Science*, 2020, *31* (10), 1302–1314.

_ , Joseph P. Simmons, and Cade Massey, "Algorithm aversion: People erroneously avoid algorithms after seeing them err.," *Journal of Experimental Psychology: General*, 2015, *144* (1), 114–126.

_ , _ , and _ , "Overcoming Algorithm Aversion: People Will Use Imperfect Algorithms If They Can (Even Slightly) Modify Them," *Management Science*, 2018, *64* (3), 1155–1170.

Erat, Sanjiv, "Avoiding Lying: The Case of Delegated Deception," *Journal of Economic Behavior & Organization*, 2013, *93*, 273–278.

European Union, "Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)," Official Journal of the European Union 2024. VERIFY citation form.

Feier, Till, Jan Gogoll, and Matthias Uhl, "Hiding Behind Machines: Artificial Agents May Help to Evade Punishment," *Science and Engineering Ethics*, April 2022, *28* (2), 19.

Fershtman, Chaim and Uri Gneezy, "Strategic Delegation: An Experiment," *The RAND Journal of Economics*, 2001, *32* (2), 352–368.

Fiorina, Morris P., "Legislator Uncertainty, Legislative Control, and the Delegation of Legislative Power," *Journal of Law, Economics, & Organization*, 1986, *2* (1), 33–51.

Freer, Mikhail, Daniel Friedman, and Simon Weidenholzer, "Motives for Delegating Financial Decisions," April 2024. arXiv:2309.03419 [econ.GN], v3 15 April 2024. University of Essex / UCSC.

Gogoll, Jan and Matthias Uhl, "Rage against the machine: Automation in the moral domain," *Journal of Behavioral and Experimental Economics*, June 2018, *74*, 97–103.

Hamman, John R, George Loewenstein, and Roberto A Weber, "Self-Interest through Delegation: An Additional Rationale for the Principal-Agent Relationship," *American Economic Review*, September 2010, *100* (4), 1826–1846.

Hill, Adam, "Does Delegation Undermine Accountability? Experimental Evidence on the Relationship Between Blame Shifting and Control," *Journal of Empirical Legal Studies*, June 2015, *12* (2), 311–339.

Holt, Charles A. and Susan K. Laury, "Risk Aversion and Incentive Effects," *American Economic Review*, 2002, *92* (5), 1644–1655.

Hüholt, Nicola and Nora Szech, "Trusting Machines with Morality – Delegating Moral Decisions to AI," *European Economic Review*, 2026, *184*, 105255.

Ivanova-Stenzel, Radosveta and Michel Tolksdorf, "Measuring preferences for algorithms — How willing are people to cede control to algorithms?," *Journal of Behavioral and Experimental Economics*, 2024, *112*, 102270.

Jauernig, Johanna, Matthias Uhl, and Gari Walkowitz, "People Prefer Moral Discretion to Algorithms: Algorithm Aversion Beyond Intransparency," *Philosophy & Technology*, 2022, *35* (1), 2.

Kirchkamp, Oliver and Christina Strobel, "Sharing responsibility with a machine," *Journal of Behavioral and Experimental Economics*, June 2019, *80*, 25–33.

Köbis, Nils, Zoe Rahwan, Raluca Rilla, Bramantyo Ibrahim Supriyatno, Clara Bersch, Tamer Ajaj, Jean-François Bonnefon, and Iyad Rahwan, "Delegation to artificial intelligence can increase dishonest behaviour," *Nature*, 2025, *646* (8083), 126–134.

Litterscheidt, Roman and David J. Streich, "Financial Education and Digital Asset Management: What's in the Black Box?," *Journal of Behavioral and Experimental Economics*, 2020, *87*, 101573.

Liu, Bingjie, "In AI we trust? Effects of agency locus and transparency on uncertainty reduction in human–AI interaction," *Journal of Computer-Mediated Communication*, 2021, *26* (6), 384–402. VERIFY: original bib listed Liu & Du as authors with a different title. No co-author Du Yu found in any indexed publication. The above is the most likely intended Liu (2021) paper; CONFIRM with Felix.

Logg, Jennifer M., Julia A. Minson, and Don A. Moore, "Algorithm appreciation: People prefer algorithmic to human judgment," *Organizational Behavior and Human Decision Processes*, March 2019, *151*, 90–103.

Maasland, Christian and Kristina S. Weißmüller, "Blame the Machine? Insights From an Experiment on Algorithm Aversion and Blame Avoidance in Computer-Aided Human Resource Management," *Frontiers in Psychology*, May 2022, *13*, 779028.

Mahmud, Hasan, A.K.M. Najmul Islam, Syed Ishtiaque Ahmed, and Kari Smolander, "What influences algorithmic decision-making? A systematic literature review on algorithm aversion," *Technological Forecasting and Social Change*, February 2022, *175*, 121390.

Naquin, Charles E. and Terri R. Kurtzberg, "Human reactions to technological failure: How accidents rooted in technology vs. human error influence judgments of organizational accountability," *Organizational Behavior and Human Decision Processes*, 2004, *93* (2), 129–141. VERIFY: original bib listed authors in reverse order (Kurtzberg, Naquin) and a different title. The above is the only 2004 Naquin-Kurtzberg paper on technology and attributions that the verifier could locate. CONFIRM with Felix.

Niszczota, Paweł and Dániel Kaszás, "Robo-investment aversion," *PLoS ONE*, 2020, *15* (9), e0239277.

Oexl, Regine and Zachary J. Grossman, "Shifting the blame to a powerless intermediary," *Experimental Economics*, September 2013, *16* (3), 306–312.

Pezzo, Mark V. and Stephanie P. Pezzo, "Physician Evaluation After Medical Errors: Does Having a Computer Decision Aid Help or Hurt in Hindsight?," *Medical Decision Making*, 2006, *26* (1), 48–56. VERIFY.

Shank, Daniel B., Alyssa DeSanti, and Timothy Maninger, "When are Artificial Intelligence versus human agents faulted for wrongdoing? Moral attributions after individual and joint decisions," *Information, Communication & Society*, 2019, *22* (5), 648–663.

Sharan, Navya Nishith and Daniela Maria Romano, "The Effects of Personality and Locus of Control on Trust in Humans versus Artificial Intelligence," *Heliyon*, 2020, *6* (8), e04572.

Steffel, Mary, Elanor F. Williams, and Jaclyn Perrmann-Graham, "Passing the buck: Delegating choices to others to avoid responsibility and blame," *Organizational Behavior and Human Decision Processes*, July 2016, *135*, 32–44.

Stein, Jens-Peter et al., "[NOT FOUND] Don't Let Me Down: The Role of Algorithmic Complexity in Algorithm Aversion," *Working Paper*, 2020. VERIFY: claim-verifier could not locate this paper in any indexed database. Title and authors may be partially fabricated by an earlier session. Original source unknown. RECOMMEND: drop the citation in literature.tex or replace with a verified algorithm-complexity-aversion reference (e.g., Castelo, Bos & Lehmann 2019 on task-dependent algorithm aversion).

Tobia, Kevin, Aileen Nielsen, and Alexander Stremitzer, "When Does Physician Use of AI Increase Liability?," *Journal of Nuclear Medicine*, 2021, *62* (1), 17–21.

Tontrup, Stephan and Christopher Jon Sprigman, "Strategic Delegation of Moral Decisions to AI," October 2025. SSRN Working Paper No. 5696827, posted 6 October 2025.

Weaver, R. Kent, "The Politics of Blame Avoidance," *Journal of Public Policy*, 1986, *6* (4), 371–398.

Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 4: Condition balance — Player A

Variable	*Punishment*	*No-Punishment*	p-value
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with p-values from two-sided t-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

We do not separately report a balance table for Player B. Player B's punishment behavior is identified within-subject across the four scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

[Comment: We can still check whether there is balance in observables for Player B.]

A.2 Full regression results — delegation decision

A.3 Robustness — delegation decision

A.4 Determinants of Player B's chosen punishment

A.5 Robustness — Player B's chosen punishment

A.6 Punishment beliefs and task performance

In the main text we claim that Player A's punishment beliefs do not vary systematically with her own task performance, and that Player As do not expect a relatively larger punishment penalty for delegation in good-outcome states than in bad-outcome states. This appendix documents the supporting analyses, restricted to the *Punishment* condition and attention-check passers ($n = 61$) for consistency with the main within-subject regression.

Table 5: Determinants of the delegation decision — full controls

	Dependent variable: *Delegation*				
	Full sample				*Passers*
	(1)	(2)	(3)	(4)	(5)
No-Punishment indicator	0.677**	0.696**	0.757**	0.721**	0.834**
	(0.320)	(0.329)	(0.334)	(0.333)	(0.376)
Task performance			−0.192		−0.254*
			(0.128)		(0.142)
Perceived performance				−0.118	
				(0.094)	
Age		−0.004	−0.004	−0.003	0.002
		(0.015)	(0.015)	(0.015)	(0.017)
Female		0.279	0.350	0.332	0.651
		(0.366)	(0.369)	(0.372)	(0.414)
Socio-economic status		−0.062	−0.058	−0.051	−0.134
		(0.113)	(0.116)	(0.115)	(0.130)
Went to uni		0.061	0.091	0.047	0.184
		(0.369)	(0.364)	(0.367)	(0.400)
Technology score		−0.110	−0.078	−0.101	0.021
		(0.149)	(0.153)	(0.148)	(0.174)
Leadership position		0.647*	0.676*	0.641*	0.413
		(0.364)	(0.367)	(0.365)	(0.412)
Constant	−0.302	0.036	0.402	0.449	0.241
	(0.226)	(0.862)	(0.930)	(0.945)	(1.062)
AME of No-Punishment (pp)	16.4**	16.5**	17.7**	16.9**	19.1**
	($p = 0.025$)	($p = 0.026$)	($p = 0.016$)	($p = 0.022$)	($p = 0.017$)
N	161	159	159	159	130
Pseudo R^2	0.020	0.039	0.049	0.047	0.063

Note: Coefficients from a Logit regression of the delegation decision on the No-Punishment indicator and the controls listed in each column. Robust (HC1) standard errors in parentheses. Significance: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$ (two-sided). Two subjects are dropped in Columns (2)–(4) due to missing socio-demographic data. The *AME of No-Punishment* row reports the average marginal effect of the No-Punishment indicator on the probability of delegation, in percentage points.

Table 6: Determinants of the delegation decision — robustness

	Dependent variable: *Delegation*		
	Full sample		*Punishment* (no excl.)
	(1)	(2)	(3)
No-Punishment indicator	0.771	0.677	
	(0.336)	(0.320)	
Task performance	−0.277		−0.395
	(0.149)		(0.185)
Overconfidence	−0.103		
	(0.094)		
Punishment beliefs: bad outcome (no del. − del.)			−0.277
			(0.499)
Punishment beliefs: good outcome (no del. − del.)			0.812
			(0.529)
Controls	yes	no	yes
Attention-check excl.	yes (Player A)	n/a	no
Sample	Full	Full	Punishment
N	159	161	78
Pseudo R^2	0.055	0.020	0.114

Note: Robustness specifications for the delegation logit. Column (1) adds *overconfidence* (defined as the weighted-average belief about own performance minus actual performance) to the main full-sample specification. The treatment effect remains: *No-Punishment indicator* $p = 0.022$. *Overconfidence* is not significantly associated with delegation $p = 0.272$. Column (2) reproduces the unconditional treatment effect for completeness ($p = 0.034$; identical to the main Col (1)). Column (3) restricts to the *Punishment* condition and includes the within-subject belief differences but *does not* exclude subjects who failed the belief-screen attention check; the heterogeneity result is preserved: *Task performance* $p = 0.033$; *Punishment beliefs: good outcome* $p = 0.125$. Robust HC1 standard errors in parentheses. Following AEA editorial policy, significance stars are not reported in the table.

Difference-of-differences in the expected delegation penalty. Define the expected punishment penalty for delegation in each outcome as $\Delta^{\text{good}} \equiv \beta^{\text{del,high}} - \beta^{\text{self,high}}$ and analogously Δ^{bad}. A within-subject paired t-test of $H_0 : \Delta^{\text{good}} = \Delta^{\text{bad}}$ yields $\bar{\Delta}^{\text{good}} - \bar{\Delta}^{\text{bad}} = +0.075$ ($t = +0.99$, $p = 0.325$) on the full Punishment sample and $+0.085$ ($t = +0.89$, $p = 0.376$) on the passers. We cannot reject equality of the two delegation penalties.

Independence of beliefs from task performance. Figure 8 plots each of the four cell-level punishment beliefs and the two outcome-conditional belief differences against Player A's task performance in the first ten rounds. Pearson correlations are reported in each panel; all six are small ($|r| \leq 0.21$) and statistically insignificant at conventional levels. In particular, the two belief differences that enter the regression in Table 3 are uncorrelated with task performance ($r = +0.12$, $p = 0.37$ for the good-outcome difference; $r = -0.15$, $p = 0.24$ for the bad-outcome difference). The within-subject correlation between beliefs and delegation reported in the main text therefore cannot be attributed to a shared dependence on task performance.

B Bonus computation

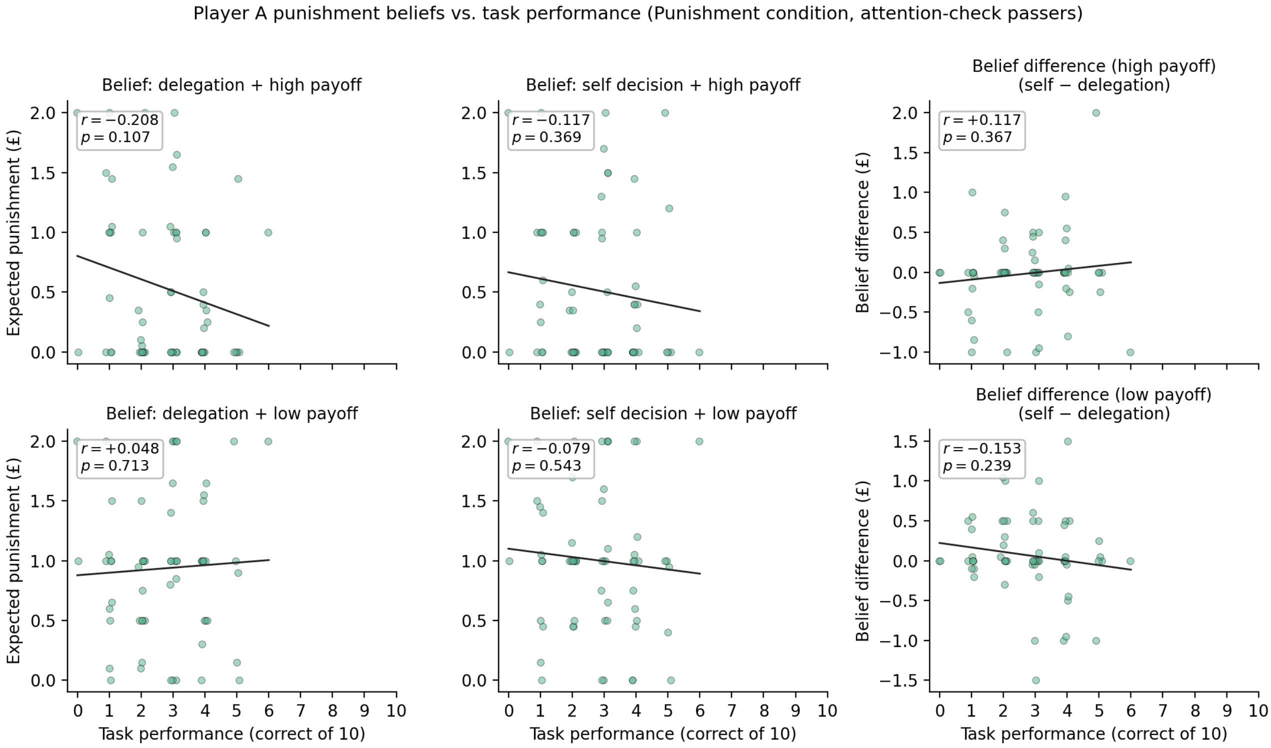
Player A spent on average 12 minutes on the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee; Player B spent on average 8 minutes and earned an

Table 7: Determinants of Player B's chosen punishment — robustness

	Dependent variable: *Punishment*	
	Excl.	*Incl. att. failers*
	(1)	(2)
Delegated	0.001	0.032
	(0.054)	(0.052)
Bad outcome	0.260	0.256
	(0.080)	(0.071)
Delegated × Bad outcome	−0.078	−0.103
	(0.052)	(0.058)
Player B belief about Player A perf. (wa)	0.001	−0.020
	(0.037)	(0.032)
Age	−0.001	0.002
	(0.006)	(0.005)
Female	−0.064	−0.100
	(0.140)	(0.128)
Socio-economic status	0.072	0.083
	(0.049)	(0.047)
Went to uni	−0.098	−0.196
	(0.142)	(0.140)
Technology score	0.030	0.073
	(0.059)	(0.053)
Leadership position	0.061	0.083
	(0.164)	(0.142)
Constant	0.020	−0.077
	(0.374)	(0.328)
Attention-check excl.	yes	no
N (subject-by-cell)	268	320
Adj. R^2	0.030	0.077

Note: Punishment regression with Player B's belief about Player A's performance (*wa_difficulty*) added as a regressor. Column (1) excludes Player Bs who failed the punishment-screen attention check (preregistered); Column (2) keeps them. Standard errors clustered at the Player B level. The delegation, outcome, and interaction coefficients are stable across both specifications, and the belief regressor is not significantly associated with chosen punishment. H2 paired-test on the bad-outcome cell: with exclusion paired-t $p = 0.153$, Wilcoxon $p = 0.533$ ($n = 67$); without exclusion paired-t $p = 0.131$, Wilcoxon $p = 0.544$ ($n = 80$). Following AEA editorial policy, significance stars are not reported.

Figure 8: Player A punishment beliefs are independent of task performance



Note: Each panel plots one belief variable against Player A's score in the first ten rounds, restricted to the *Punishment* condition and attention-check passers ($n = 61$). Points are jittered horizontally for legibility. Lines are OLS fits; annotations report Pearson r and its two-sided p-value. Top row: good-outcome beliefs (delegation, self, and the difference *self* − *delegation*). Bottom row: low-outcome beliefs.

average bonus of £2.30. Bonuses are computed deterministically from the cleaned data per the experimental rules: random-round task payoff (expectation over the uniform draw across rounds 1–10); belief-about-own-score payoff (£0.30 times the probability mass placed on the realized score); belief-about-distribution payoff (expectation over the uniform random draw of which score is checked, paying £0.30 if the estimate is within 5 percentage points of the true population share); and—for Player A in the *Punishment* condition—expected punishment from the matched Player B's strategy-method choice in the realized cell, with £0.10 per cell of strategy-method punishment cost for Player B. The reported bonuses exclude the risk-elicitation MPL payoff (the raw-data column was not retained in the cleaned files); including it would add roughly £0.30–£0.50 per subject.

C Ex-ante power calculations

The cell size of 80 subjects per role per condition was set to detect, at conventional power, the effect sizes anticipated from the responsibility-avoidance literature. We report the minimum detectable effect (MDE) for each preregistered test below, computed at $\alpha = 0.05$ (two-sided) and power $= 0.80$. For H1, the relevant test is a two-sample z-test of proportions (delegation rate in *Punishment* vs *No-Punishment*); the MDE is approximately invariant to the assumed baseline rate. For H2, the relevant test is a paired-mean-difference test on Player B's punishment in the bad-outcome cell; because the within-subject standard deviation σ_d of the punishment difference is unknown ex ante, we report the MDE for a range of values of σ_d expressed as multiples of the equivalence bound used in Result 2 of the body of the paper ($\delta_a = £0.20$, equal to 10% of the maximum punishment of £2).

Table 8: Minimum detectable effects under the preregistered design

Hypothesis	Assumption	Minimum detectable effect
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.30$	$\Delta = 0.216$ (21.6 pp)
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.50$	$\Delta = 0.215$ (21.5 pp)
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.70$	$\Delta = 0.179$ (17.9 pp)
H2 (paired, within-subject)	$\sigma_d = 0.5\,\delta_a = £0.10$	$\Delta = £0.031$
H2 (paired, within-subject)	$\sigma_d = 1\,\delta_a = £0.20$	$\Delta = £0.063$
H2 (paired, within-subject)	$\sigma_d = 2\,\delta_a = £0.40$	$\Delta = £0.125$
H2 (paired, within-subject)	$\sigma_d = 3\,\delta_a = £0.60$	$\Delta = £0.188$
H2 (paired, within-subject)	$\sigma_d = 4\,\delta_a = £0.80$	$\Delta = £0.251$

Note: Both tests assume $n = 80$ per role per condition, $\alpha = 0.05$ two-sided, power $= 0.80$. For H2, σ_d is the within-subject standard deviation of the punishment difference (delegated minus non-delegated, conditional on bad outcome) and is expressed as a multiple of the anchor effect $\delta_a = £0.20$ (the equivalence bound used in Result 2 in the body of the paper, equal to 10% of the maximum punishment of £2). Δ is the corresponding minimum detectable effect in pounds.

D Preregistration and deviations

The experiment was preregistered at AsPredicted #133980, available at aspredicted.org/hs9az8.pdf and registered on 30 May 2023 prior to the main wave of data collection. The preregistration commits to the two research questions stated in the body as H1 and H2, the sample-size target (320 subjects total, 80 per role per condition), the desktop-only restriction, the use of the strategy method to elicit Player B's punishment, and a set of primary statistical tests (parametric t-tests and non-parametric MWU and Kolmogorov–Smirnov tests for both comparisons; probit regressions for the delegation decision and linear regressions for punishment behavior with treatment, beliefs, and socio-demographics as predictors).

The preregistration also notes that a pilot of 25 observations had been collected in the *Punishment* treatment prior to registration; these observations are pooled with the main wave in all analyses, as preregistered. Two secondary analyses are pre-stated in the registry: the role of own-performance and beliefs about own performance (developed as M2 in Section 4.1), and the consistency of Player A's beliefs about expected punishment with her delegation decision (developed as M3 in Section 4.1).

The table below summarizes every analysis reported in the body of the paper, the corresponding preregistered specification, and any deviation.

The full set of preregistered tests, the deviations from each, and the rationale appears in Appendix D

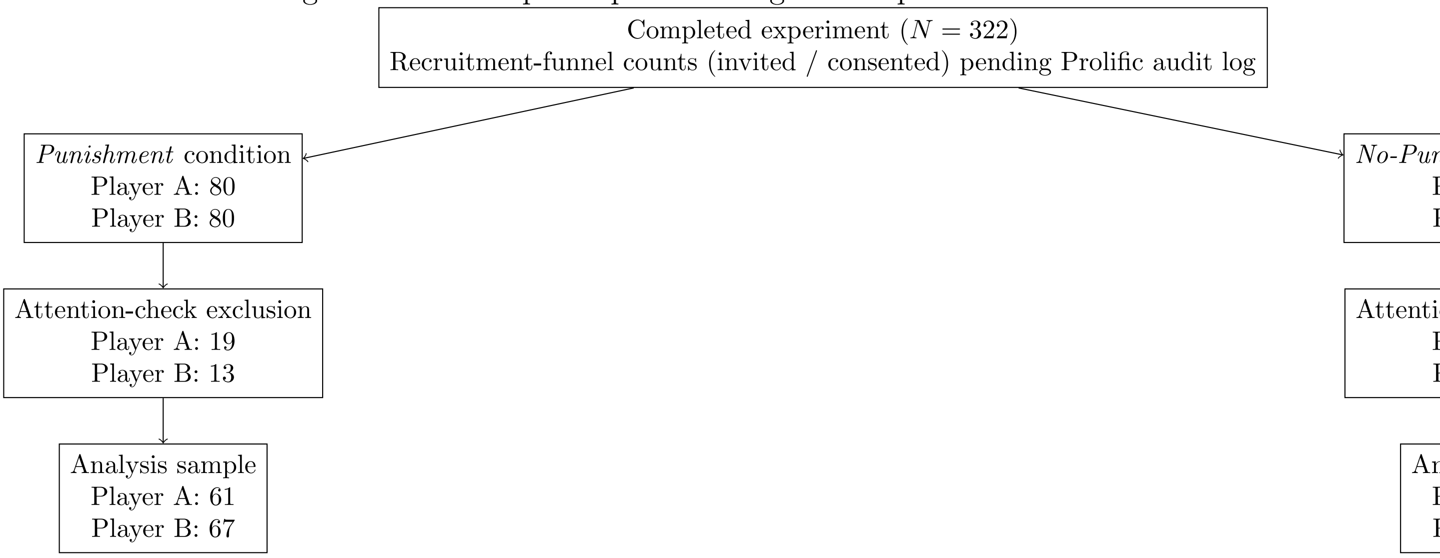
Table 9: Preregistered tests, reported tests, and deviations

Preregistered specification	As reported	Deviation / rationale
H1: between-treatment test of delegation rates (*Punishment* vs *No-Punishment*); preregistered tests are parametric t-test, MWU, KS. Direction not specified in the registry.	Pearson Chi-squared and Fisher's exact, two-sided ($p = 0.033$ / $p = 0.041$).	Test family substituted (Chi-squared and Fisher's exact are the standard tests for a binary outcome; the preregistered t-test, MWU and KS tests on a binary outcome are equivalent to the proportion test). The realized direction is opposite to that anticipated by the responsibility-avoidance literature on which we drew.
H2: paired test of Player B punishment in the bad-outcome cell, delegated vs non-delegated, attention-check passers only	Paired t-test ($p = 0.153$); Wilcoxon signed-rank ($p = 0.533$).	As preregistered.
Probit / logit of delegation on condition + controls (Table 1, Cols (1)–(2))	Logit, as reported.	As preregistered. (Probit and logit are equivalent for our purposes; we report logit for ease of marginal-effect interpretation.)
Logit on *Punishment* subsample with belief differences (Col (3))	As reported.	Specification preregistered; sample restriction follows the preregistered attention-check rule.
Pre-stated secondary analysis: own-performance and beliefs about own performance	Reported as M2 in Section 4.1.	As preregistered (secondary).
Pre-stated secondary analysis: beliefs about expected punishment vs delegation decision	Reported as M3 in Section 4.1.	As preregistered (secondary).
Mechanism analyses M1 (process ownership) and M4 (experimenter demand) (Section 4.1)	As reported.	*Not preregistered.* Treated as exploratory.

E Flow of participants

The flow of participants through the experiment, by condition and role, is shown in Figure 9. Recruitment-funnel counts (invited / consented) are pending the Prolific audit log.

Figure 9: Flow of participants through the experiment



A flow diagram showing recruitment, attrition, attention-check exclusion, and the analysis sample, broken down by condition and role, appears in Appendix E; in particular, attrition rates do not differ meaningfully across the *Punishment* and *No-Punishment* conditions.

We preregistered exclusion of subjects who failed the attention check on the screen at which their main outcome variable was elicited. For Player A, this is the belief-elicitation screen; for Player B, the punishment-elicitation screen. Of 161 Player Bs, 135 pass the attention check (83.9%); the corresponding pass rate for Player A is 131/161 (81.4%). All main results are robust to including subjects who fail the attention check; we report these robustness checks in Appendix A.2.

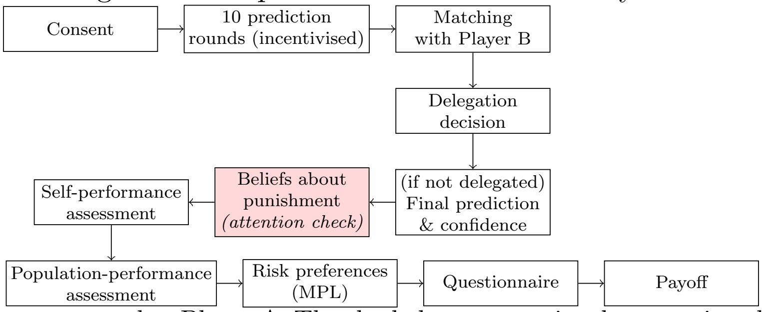
FB: maybe move to first set of results. Also the appendix does not show anything and the default should be including everyone.

F Additional Figures

F.1 Experimental timelines

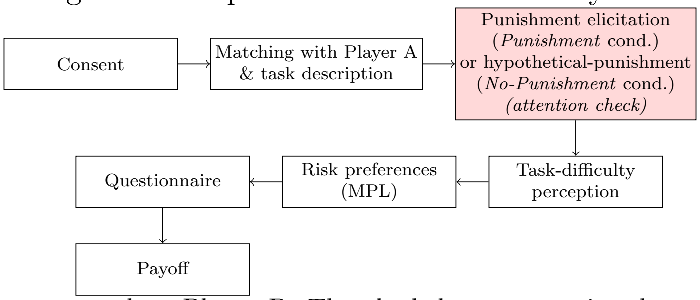
F.2 Punishment frequencies

Figure 10: Experimental timeline — Player A



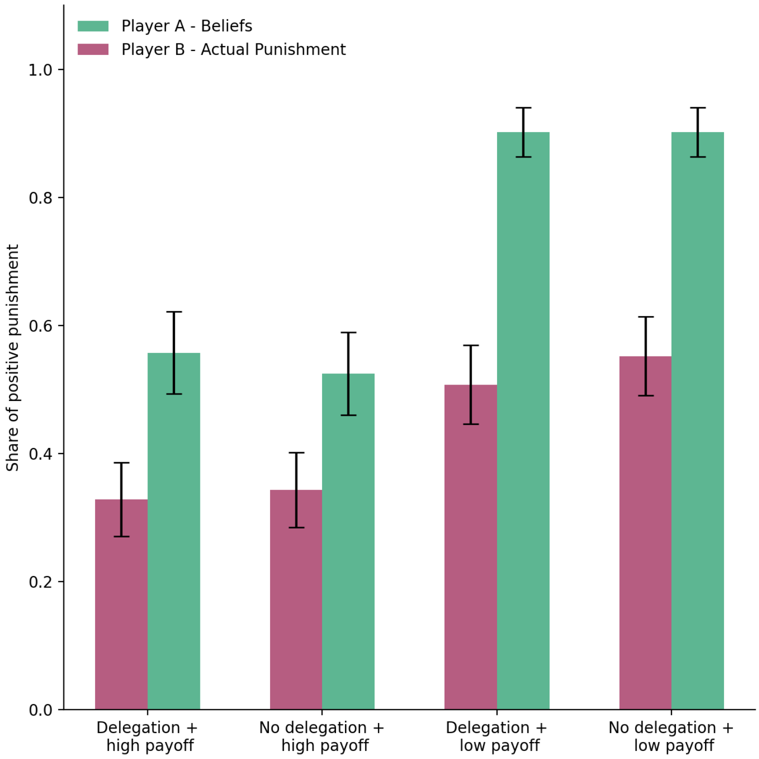
Note: Screen sequence presented to Player A. The shaded screen carries the attention check used to identify subjects excluded from belief-related analyses. The final-prediction-and-confidence screen is skipped if Player A delegates to the algorithm.

Figure 11: Experimental timeline — Player B



Note: Screen sequence presented to Player B. The shaded screen carries the attention check used to identify subjects excluded from punishment-related analyses. In the *No-Punishment* condition, no actual punishment is implemented; the elicitation is hypothetical and serves to maintain comparability of the screen sequence across conditions.

Figure 12: Frequencies of punishment and beliefs about punishment



Note: Share of Player Bs choosing strictly positive punishment in each of the four scenarios (solid bars) and share of Player As assigning a strictly positive expected punishment in each scenario (hollow bars).